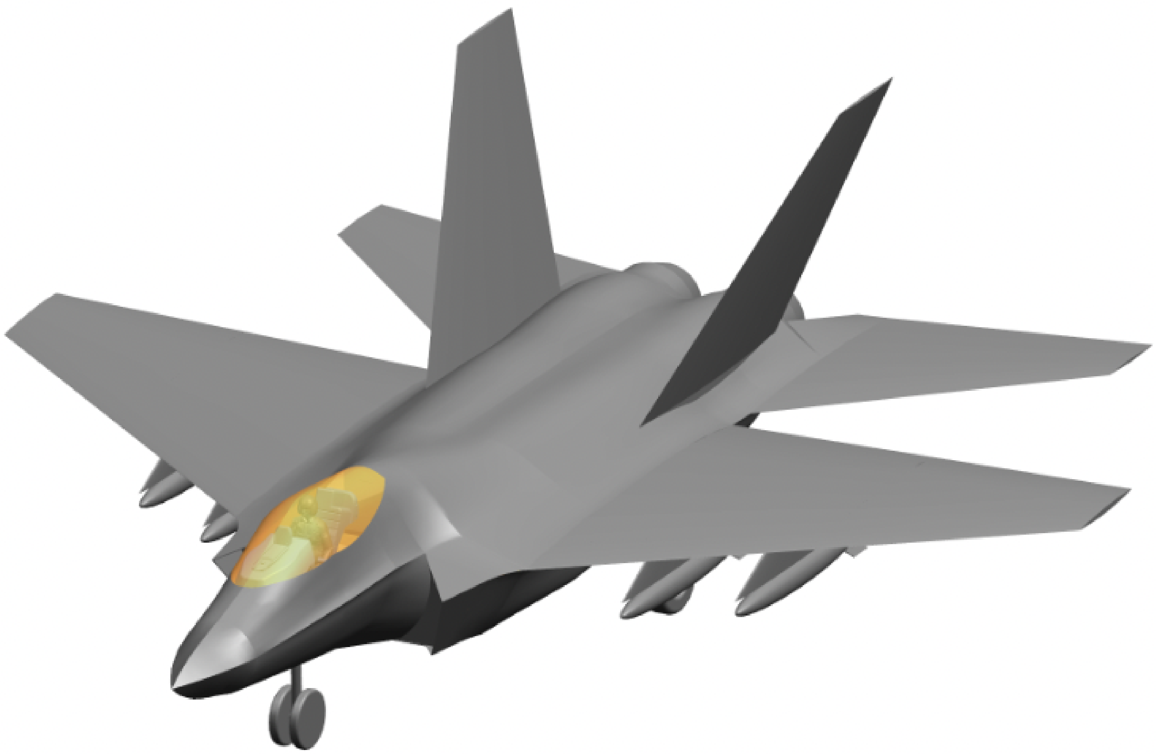
F/A-67 CDR Report
AE 481 Team 01 - No Thai Grubbin



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1 Summary

This document presents the Critical Design Review (CDR) report for Team 01, No Thai Grubbin, and their entry for the Next Generation Carrier-Based Strike Fighter (NGCSF) competition, the F/A-67 Condor. This report covers the conceptual design of the aircraft, including design decisions, analyses, and mission performance.

1.1 Executive Summary

Combat aircraft cost has been consistently increasing over time, as described by Augustine's Law. The United States Navy requires this trend to be broken to maintain the number of aircraft in its inventory. The American Institute of Aeronautics and Astronautics (AIAA) published the NGCSF Request for Proposal (RFP), seeking a strike fighter aircraft replacement for the F/A-18E/F naval aircraft. The outlined objective is to deliver a strike fighter that provides a credible combat capability that exceeds the performance of legacy aircraft, while maintaining a comparable unit cost.

The RFP outlines two missions, air-to-air and strike, with several mission performance requirements. This includes a combat radius, combat time, dash speeds, sustained turn rate, and vertical load factor. Additionally, ordnance for each mission is outlined. The performance objective is derived from mission requirements, with the overall goal to increase performance values from their minimum to their desired values. Preliminary sizing of the aircraft relied on various historical regressions to provide estimations of takeoff weight, with mission performance parameters as inputs. Initial design points were verified using thrust-to-weight (T/W) vs. wing loading (W/S) constraints and thrust vs. wing area sizing plots. Initial trades included the selection of the number of engines and crew. After the Preliminary Design Review (PDR), several refinements were made to sizing models, including empty weight prediction, parasitic drag estimation, and fuel burn analysis. Additional trade studies were performed to facilitate the reselection of a design point, as well as to determine the tradeoffs between internal and external ordnance carriage. For both PDR and CDR, design points were selected to increase mission performance beyond the minimum requirements, while maintaining the unit cost requirement, thus satisfying the objective outlined by the RFP. The design of aerodynamic surfaces, the outer mold line, and landing gear were subsequently performed.

The design developed by No Thai Grubbin, the F/A-67 Condor, meets and exceeds the RFP mission and performance requirements, with a twin engine, single crew, carrier-based strike fighter. The aircraft has a takeoff weight of 60,826 lb with an empty weight of 31,207 lb. The aircraft is equipped with two F110-GE-132 engines, producing a maximum installed thrust of 59,568 lb. The combat radius meets the desired 1,000 nmi for both the air-to-air and strike missions. It is capable of a supersonic dash of Mach 1.65 at 30,000 ft, and Mach 0.9 at sea-level. The aircraft has a swept wing with leading edge flaps and trailing edge flaperons, an all-moving horizontal stabilator, and twin canted vertical stabilizers, with rudders. Additionally, the aircraft has complete internal storage of ordnance for both missions as requested by the RFP. The No Thai Grubbin team is confident in the F/A-67 design based on the analysis and technical justification presented in this report.

1.2 Design Drawings

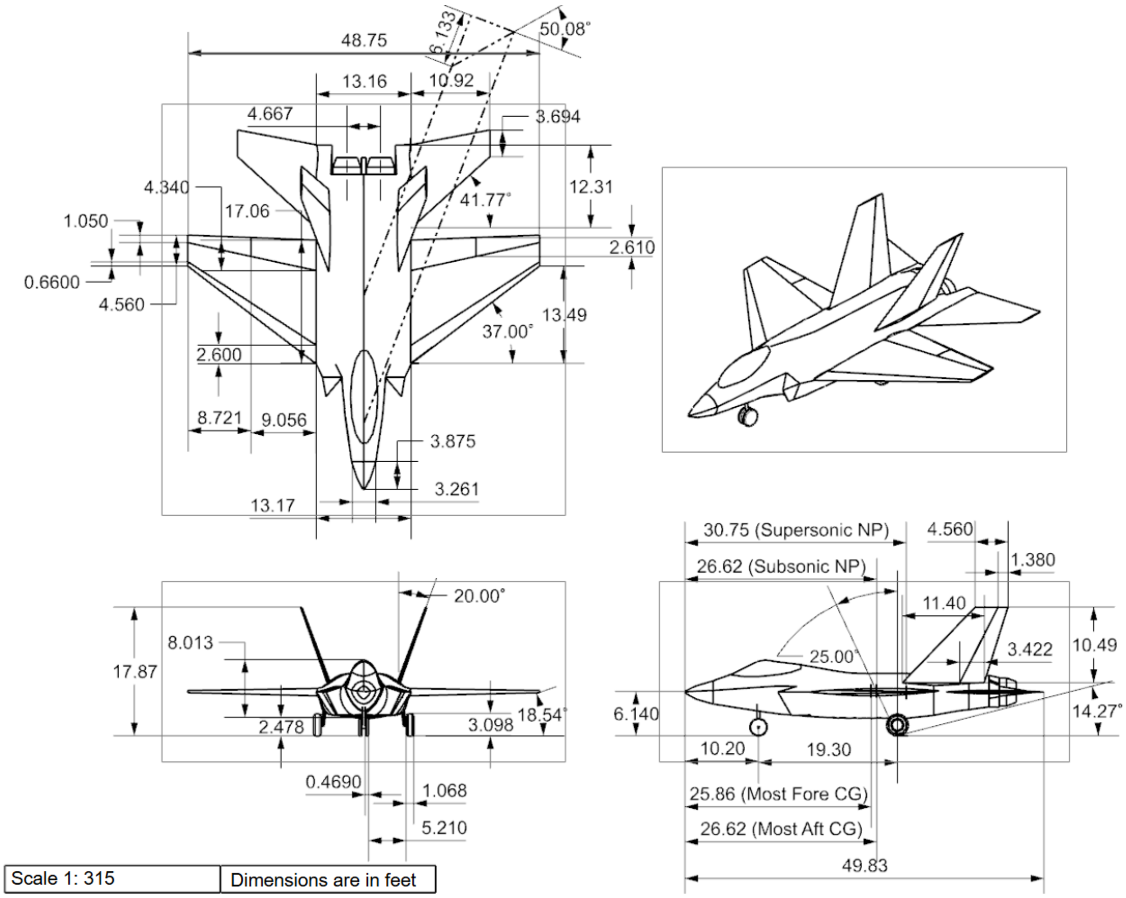

Figure 1: Dimensioned three-view drawing of the F/A-67.

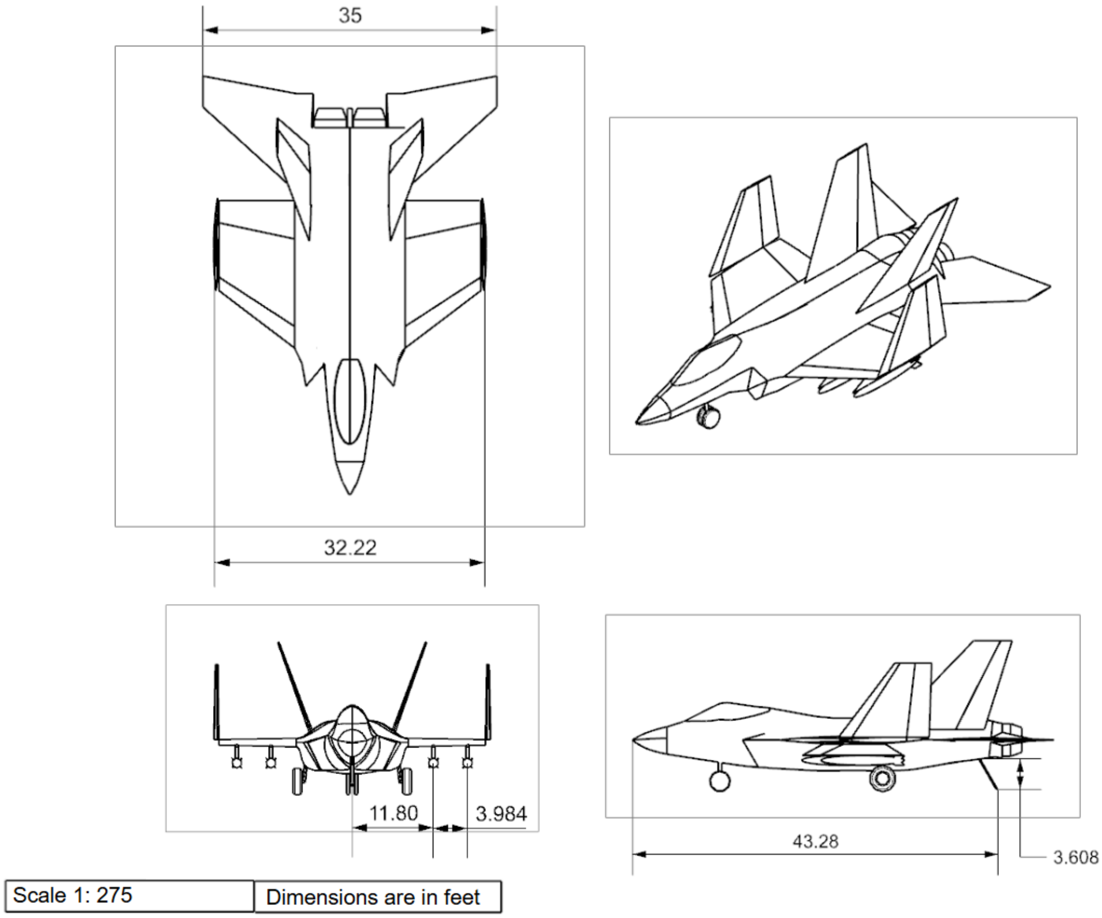

Figure 2: Stowed three-view drawing of the F/A-67.

1.3 Aircraft Geometry and Performance Parameters

Geometric and performance parameters are presented here for the F/A-67, as well as several similar combat aircraft. All costs shown are reported in 2025 dollars.

Table 1: Aircraft comparison table

	F/A-67	F/A-18E	F-35C	F-15E
Takeoff Weight (lb)	60826	66000	70000	81000
Empty Weight (lb)	31207	32081	34800	37500
Air-to-Air Payload (lb)	2460	2946	up to 18000	2460
Strike Payload (lb)	4424	5442	up to 18000	4398
Combat Radius (nmi)	1000	444-462	670	687
Air-to-Air Fuel Burn (lb)	26959	18218	18600	22300
Strike Fuel Burn (lb)	21425	18313	18600	22300
Cruise/Landing/Takeoff C_L	0.42/0.93/1.2	-	-	-
Cruise L/D	11.7	-	-	-
T/W	0.9890	0.651	0.614	0.728
Wing Loading (lb/ft^2)	94	132	105	133
Engine Type	F110-GE-132	F414-GE-400	F135-PW-400	F110-GE-129
Maximum SLS Static Thrust	33093	21496	43000	29474
TSFC (dry)	0.68	0.82	-	0.67
Span (ft)	48.75	44.7	43	42.8
Reference Area (ft^2)	648	500	668	608
Aspect Ratio	3,67	3.5	2.7	3.0
Average wing t/c	5.7%	-	-	-
Maximum Cruise Mach Number	1.8	1.6	1.6	2.5
Economic Cruise Mach Number	0.84	0.84	-	-
Subsonic Static Margin	0.86%	-	-	-
Supersonic Static Margin	28%	-	-	-
Maximum Landing Distance (ft)	6800	8500	-	12,000
Maximum Takeoff Distance (ft)	4800	4666	-	11000
Spot Factor	1.07	1	-	-
Flyaway cost for 500 units ($M)	98.8	113	127	61.4

2 Introduction

This section summarizes the mission and design requirements outlined by the NGCSF RFP.

2.1 Project Objective

The primary purpose of the NGCSF competition is to deliver a replacement aircraft for the F/A-18E/F, with a credible combat capability and comparable unit cost. This has been interpreted as designing an aircraft with a unit flyaway cost less than the F/A-18E/F, while attempting to maximize the mission performance parameters. The target initial operating capability date is 2035, for a production run of 500 aircraft.

General design requirements are specified, such as only allowing existing production engines, currently available materials in production quantities, and Technology Readiness Level (TRL) 6 or higher subsystem technologies. Additionally, internal carriage of ordnance is desired for reduced radar cross section. Finally, aircraft compatibility with existing aircraft carriers is required.

2.2 Missions

The RFP outlines two missions, air-to-air and strike, both with a set of performance requirements. Additional carrier suitability requirements for carrier launch and recovery performance, and point performance requirements are further specified and apply to both missions.

2.3 Air-to-Air

Air-to-air combat is interpreted as a fighter escort mission profile as outlined in the Standard Aircraft Characteristics document of the F/A-18E [1]. The performance requirements are as follows:

- Minimum 700 nmi combat radius, desired 1000 nmi
- Minimum 2 min combat at maximum thrust and best turning speed at 10,000 ft, desired 5 min
- Ordnance carriage for the entire mission
- Minimum 8 deg/s sustained turn rate at 20,000 ft and mid-mission fuel weight, desired 10 deg/s
- Minimum Mach 1.6 dash speed at 30,000 ft, desired Mach 2.0

The air-to-air mission includes a climb and cruise out from the carrier before descending to combat. After combat, the aircraft climbs and cruises back to the carrier. The mission profile is illustrated in Figure 3. Credit for distance traveled was taken for climb and cruise segments, with no credit taken for descents, combat, and loiter. Additionally, the cruise segments were assumed to be cruise-climbs at a constant Mach number of 0.84. This is in line with specifications in [1]. The RFP does not specify where the supersonic dash occurs, so it is assumed to occur before combat.

2.4 Strike

The strike mission profile is interpreted as an interdiction mission where combat involves sea-level dashes into and out of the combat area. The performance requirements are:

- Minimum 700 nmi combat radius, desired 1000 nmi
- 50 nmi sea-level dashes (ingress and egress) at intermediate thrust
- Minimum Mach 0.85 dash speed at intermediate thrust, desired Mach 0.9
- Ordnance carriage for the entire mission

The strike mission follows a similar profile to the air-to-air mission, with the only difference being the combat segments. Additional details on assumptions made on the mission profiles for takeoff weight estimation and their effects on the sizing process are covered in §4 and §14. The mission profiles are shown in Figure 3.

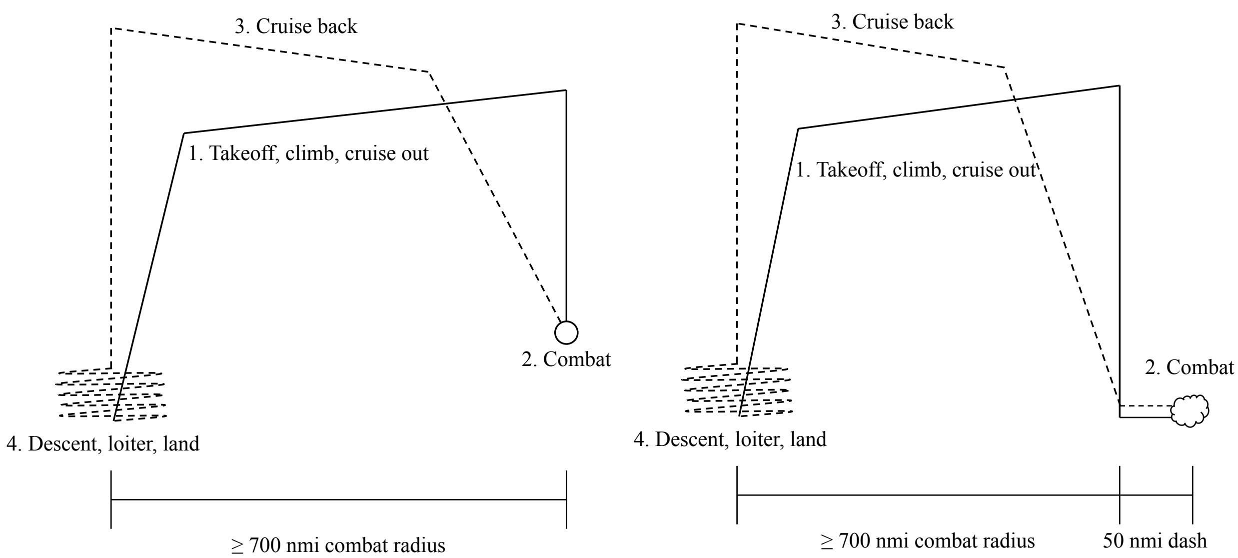

Figure 3: Mission profiles for the air-to-air (left), and strike (right) missions, as outlined in the RFP.

2.5 Carrier Suitability Requirements

In addition to requiring compatibility with existing aircraft carriers, the RFP lists several requirements for launch and recovery performance. They are evaluated during the preliminary sizing of the aircraft as part of both mission profiles.

Launch is evaluated at tropical day conditions with zero wind over deck (WOD). The aircraft is required to have a minimum catapult end speed as the highest of several conditions including altitude loss, lift limit, pitch rate limit, and minimum longitudinal acceleration.

Recovery requires arrestment engaging speed to be 5% greater than approach speed, which is 10% greater than stall speed. Additionally, approach speed must be less than 145 kts. Maximum landing weight includes 25% of the maximum fuel weight and 50% store weight. The fuel weight shall be sufficient for two landing attempts and a 20 min loiter at 10,000 ft. Point performance requirements specify a maximum WOD of 15 kts. The mission profiles include two landing attempts, and 20 min loiters for fuel burn estimation. Reserve fuel is assumed to be 5% of mission fuel weight [1].

2.6 Point Performance Requirements

Further requirements for aircraft performance include a minimum 7g vertical load factor with a desired value of 8g at mid-mission weight. For multi-engine designs, a single-engine rate of climb (SEROC) is specified at 200 ft/min at launch and 500 ft/min for approach. Dimensional constraints for the aircraft include a maximum length of 50 ft, height of 18.5 ft, unfolded wingspan of 60 ft, and folded wingspan of 35 ft. A provision is made for minimizing the spot factor of the aircraft. Finally, the maximum takeoff weight of the aircraft shall not exceed 90,000 lb.

3 Configuration

3.1 Preliminary Configuration

The preliminary configuration of the F/A-67 revolved around analyzing preliminary sizing trends and design drivers, based on results using the plots and processes outlined in §4. The conclusions from the preliminary sizing analysis are as follows:

Increasing mission performance increases takeoff weight. For example, increasing combat radius, combat time, and dash Mach number for the air-to-air mission all increase the takeoff weight. High takeoff weights result in infeasible designs due to limitations by the catapult launch requirements and therefore is a design driver. This occurs before the aircraft reaches the RFP 90,000 lb limit. Therefore, an upper limit for mission performance parameters exists. Slotted flaps were selected to increase $C_{L,\max}$ which relax the catapult launch constraint, increasing the maximum feasible takeoff weight.

Increasing aspect ratio (AR) increases lift-to-drag ratio (L/D) which reduces cruise fuel burn. This has the effect of reducing takeoff weight for a given set of mission performance parameters. Alternatively, this can result in an increased mission performance for a given takeoff weight. However, increasing AR requires increasing wingspan or reducing wing reference area. The RFP aircraft dimensions place an upper bound on aircraft wingspan, length, and height. Wingspan was maximized to the 60 ft limit, while wing area was minimized to reduce aircraft size and weight. This enables higher mission performance while satisfying the dimensional constraints.

The air-to-air dash requirement and takeoff constraints set a lower bound on aircraft thrust. The number of engines was set at 2 to provide sufficient thrust and enable an increased air-to-air dash speed.

Design decisions for the configuration of the F/A-67 were determined to maximize mission performance. This was done in conjunction with the unit cost constraint set by the RFP. The rest of this report details how further design decisions were made, such as the aerodynamic configuration, fuselage layout, and engine selection.

The preliminary design of the F/A-67 is sized for a combat radius of 900 nmi and 1,020 nmi for the air-to-air and strike missions respectively. For the air-to-air mission, the aircraft has a combat time of 2.5 min with a 11 deg/s sustained turn rate. Furthermore, the F/A-67 is capable of a Mach 1.7 supersonic dash at 30,000 ft and Mach 0.9 at sea-level. The aircraft can also pull an 8g vertical load factor. All of the mission performance parameters exceed the minimum requirements, with several meeting or exceeding the desired values.

3.2 Critical Configuration

After reviewing feedback from PDR, several changes were made to the configuration process. One major point of concern was the preliminary F/A-67 having a lower unit cost than the F/A-18E/F, despite having a larger takeoff weight. Configuration decisions were revised to put a larger emphasis on cost reduction. This includes not exceeding desired mission performance parameters when possible. Certain performance parameters that fell out of the analysis, such as sustained turn rate or air-to-air combat best turning speed, were allowed to exceed desired values. This is because they were largely driven by other mission performance parameters like the supersonic dash requirement. For example, the minimum supersonic dash Mach number of 1.6 requires a higher T/W than the desired 10 deg/s sustained turn rate requirement. Sustained turn rate is therefore considered to be a "fall out" parameter. The requirement that air-to-air combat occurs at best turning speed is also a "fall out" parameter. Additionally, no minimum value for best turn rate is specified, so it is not considered in sizing plot analyses since any aircraft has a best turn speed and turn rate. The team considers the sustained turn rate requirement to satisfy maneuverability requirements for the aircraft.

The revised configuration decisions and their justification are as follows. Mission performance is increased to satisfy the objective, while taking greater consideration in how it increases takeoff weight. The catapult launch requirements are still a driver in terms of aerodynamic feasibility of a design. To reduce empty weight and cost, AR and wingspan are only as large as necessary to produce a feasible design. This differs from the methodology of maximizing wingspan and AR used for PDR. Wing area is reduced to maintain compliance with RFP dimensional constraints. In particular, the wing reference area should be less than the PDR design. This is because the preliminary design was already close to the dimension limits.

Following the new methodology, the F/A-67 configuration was revised for CDR. The wing reference area is 648 ft^2, less than the 670 ft^2 of the preliminary design. The wingspan is 48.75 ft giving an AR of 3.67. The wings include slotted trailing edge flaps and flaperons, and leading edge flaps to provide sufficient lift during catapult launch. An all-moving horizontal stabilator is used to enable high maneuverability and sufficient control during air-to-air combat. Twin, canted vertical stabilizers with rudders are used to reduce radar cross-section (RCS), and to reduce the height of the aircraft. The F/A-67 is powered by two engines to provide the necessary T/W for supersonic dash. Payload for both missions are stored internally to reduce drag and observability.

Mission performance for the air-to-air mission meets the desired 1000 nmi combat radius, achieves 3.5 min of combat time, and Mach 1.65 dash speed. The sustained turn rate exceeds the desired value and reaches 12 deg/s. This falls out of the analysis, and is driven by the dash speed. For the strike mission, the aircraft also meets the desired 1000 nmi combat radius and the desired Mach 0.9 dash speed. For both missions, the F/A-67 is aerodynamically and structurally capable of withstanding the desired 8g vertical load factor. Further justification and trade studies are discussed in greater detail in §14.

4 Preliminary Sizing

Preliminary sizing of the F/A-67 is performed using an iteration of an initial weight estimate of the aircraft, combined with performance constraints and requirements to determine the feasibility of a particular design. The inputs for a given iteration are aircraft parameters like wing reference area and engine thrust, as well as mission performance parameters like combat radius. The selection of the design point is consistent with the justification outlined in §3.

4.1 Initial Weight Estimation

Takeoff weight estimation is based off of the air-to-air and strike mission profiles outlined in the RFP using algorithms from Martins [2]. Weight fractions for individual mission segments are calculated using equations from both Martins and Nicolai [3]. Additionally, certain segments such as takeoff, descent, and landing are assumed to have constant weight fractions as listed in Roskam [4]. A key component of the initial weight estimate is the Breguet range equation (1).

$$R = \frac{V}{c}\frac{L}{D}\ln\frac{W_i}{W_{i+1}} \tag{1}$$

Specifying a range based on a desired combat radius yields the weight fraction for a particular cruise segment. Other mission segments not explicitly outlined in the mission profiles include weight fraction estimates for accelerations [3]. This includes modeling climbs as accelerations to a cruise speed, and accelerations to dash speeds before combat. A 20 min loiter mission segment is included before conducting two landing attempts. Loiter weight fractions are calculated using equation (2).

$$E = \frac{L/D}{c}\ln\frac{W_i}{W_{i+1}} \tag{2}$$

Both the range and endurance are influenced by the aircraft L/D, with increasing L/D resulting in lower fuel consumption for a given range or endurance. For maximizing range, an L/D of $0.943L/D_{\max}$ was used, based on a cruise climb described in Martins [2]. Endurance is maximized using $L/D_{\max}$. To determine $L/D_{\max}$, a quadratic drag polar is assumed. This results in an expression for $L/D_{\max}$ as

$$L/D_{\max} = \frac{1}{2}\sqrt{\frac{\pi ARe}{C_{D0}}} \tag{3}$$

Parameters such as AR and e are determined using the aircraft geometry or historical values. The parasitic drag coefficient, C_{D0} is determined using historical regressions.

Reserve fuel weight is also calculated as the sum of fuel required for a 20 min sea-level loiter and 5% of the mission fuel weight, which is consistent with statements presented in the F/A-18E Standard Aircraft Characteristics document. Trapped fuel weight is determined to be 1% of mission fuel weight. The other component needed for takeoff weight estimation is a prediction of aircraft empty weight. This is determined using regressions of empty weight fraction for fighter aircraft sourced from Raymer [5]. Other weights include payload weight and crew weight.

The preliminary weights for the F/A-67 are included in Table 2. The highest takeoff weight of 76,366 lb is within the 90,000 lb requirement listed in the RFP.

Table 2: Mission weight breakdown

	Air-to-Air	Strike
Empty Weight (lb)	41,432	41,397
Crew Weight (lb)	200	200
Payload Weight (lb)	2,460	4,424
Fuel Weight (lb)	32,274	30,270
Takeoff Weight (lb)	76,366	76,292

4.2 Thrust-to-Weight vs. Wing Loading

The initial sizing of the F/A-67 utilized a T/W-W/S sizing plot to analyze the feasibility of a particular design. Constraint curves were plotted for various flight maneuvers and for each mission. The curves are derived from Martins [2], Nicolai [3], and Raymer [5]. The equations for the constraints, as well as their operating conditions are as follows:

- Dash: 30,000 ft at Mach 1.7 (air-to-air), sea-level at Mach 0.9 (strike). The constraints are evaluated at their respective combat weight fractions.

$$\frac{T}{W} = \frac{qC_{D0}}{W/S} + \frac{W}{S}\left(\frac{K}{q}\right) \tag{4}$$

  where $q \equiv 0.5\rho V^2 = 0.5\gamma p M^2$ is the dynamic pressure, and $K \equiv 1/(\pi ARe)$

- Cruise: 40,000 ft at Mach 0.84. The operating condition is based on the cruise speed and altitude of the F/A-18E. The constraint utilizes equation (4). Each mission has two cruise segments for cruising to and from combat.

- Ceiling: 50,000 ft at Mach 0.84. The RFP does not specify a cruise ceiling, but one is listed for completeness. The constraint utilizes equation (4). Ceiling is evaluated for the cruise out segment, since it is the most constraining.

- Sustained turn rate: 10 deg/s at 20,000 ft (air-to-air). The required T/W is found using a system of equations (5). The constraint is evaluated at the air-to-air combat weight fraction.

$$\begin{aligned} n &= \sqrt{\left(\frac{\dot{\psi}V}{g}\right)^2 + 1} \\ \frac{W}{S} &= \frac{q}{n}\sqrt{\frac{C_{D_0}}{K}} \\ \frac{T}{W} &= \frac{qC_{D_0}}{W/S} + \frac{W}{S}\left(\frac{n^2K}{q}\right) \end{aligned} \tag{5}$$

- Vertical load factor: 8g at the combat condition and weight fraction. The aircraft is assumed to be flying at Mach 0.95, based on assumptions from Nicolai. The constraint is calculated using equation (6).

$$\frac{W}{S} = \frac{qC_{L,\max}}{n_z} \tag{6}$$

- Climb rate: SEROC of 200 ft/min and 500 ft/min for takeoff and approach respectively. Additional climb constraint curve are added for the 2 climbs to cruise altitude (500 ft/min) and an additional climb for the strike mission during combat (12,800 ft/min). The combat climb rate is determined using F/A-18E Standard Aircraft Characteristics data. Equation (7) calculates the T/W required for the mission segment. Additional factors are applied for the one engine inoperative condition, the use of maximum thrust, and tropical day temperatures.

$$\frac{T}{W} = \frac{V_\gamma}{V} + \frac{\rho V^2 C_{D_0}}{2(W/S)} + \frac{2K}{\rho V^2}\frac{W}{S} \tag{7}$$

- Conventional takeoff: 3,300 ft ground roll at 4,000 ft altitude. The ground roll is set to be roughly half of the shortest runway that would be typically used by the F/A-67 [6].

- Conventional landing: 6,500 ft ground roll at 4,000 ft altitude. Maximum landing weight is used as specified by the RFP.

- Catapult launch and recovery: The constraints are implemented using the RFP requirements. A full explanation of the derivation of the constraint equations for takeoff, landing, catapult launch, and recovery can be found in Appendix B. These equations depend on $C_{L,\max}$ and are 1g maneuvers, which satisfies the need for a stall constraint in the sizing plots.

All equations take for input either the T/W or W/S at their specific point in the mission profile. The output values are then converted back to takeoff values. This ensures that all constraint curves can be plotted against the takeoff design point.

The preliminary sizing plots for the F/A-67 are shown in Figures 4 and 5. Constraints are plotted as dashed or solid lines for maximum thrust or intermediate thrust respectively. For instance, the air-to-air dash constraint is a maximum thrust constraint, while for strike, it is an intermediate thrust constraint. Design points are plotted for both the maximum T/W and intermediate T/W, with feasible regions highlighted in green and yellow respectively. The preliminary F/A-67 has a takeoff W/S of 114 lb/ft^2 and a maximum T/W of 0.78 and 0.42 for maximum and intermediate thrust respectively. Based on the sizing plots, the F/A-67 is within the feasible region for all constraints.

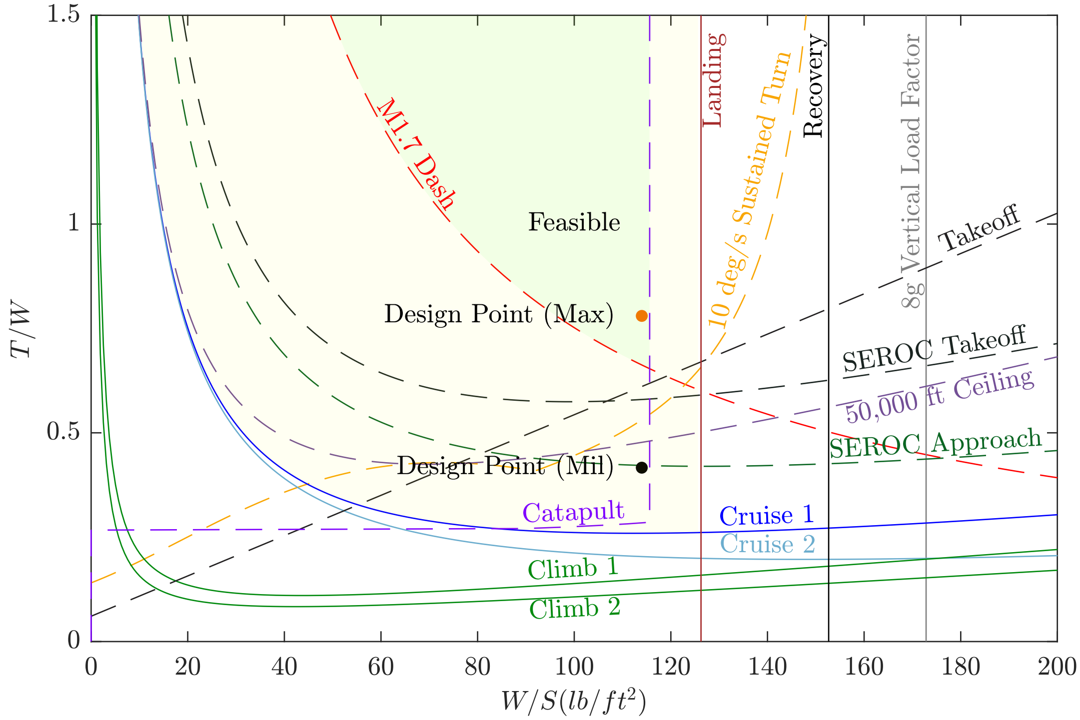

Figure 4: T/W-W/S plot for the air-to-air mission.

Early sizing plot trades found that the air-to-air mission was more constraining then the strike mission. This was largely due to the increased fuel requirements to perform the combat mission segment. The air-to-air mission requires maximum thrust during combat, while strike only mandates intermediate thrust. It follows that increasing the combat time increases the combat fuel needed, which cascades to increasing the total fuel for the entire mission. Therefore, the air-to-air mission has to balance combat radius, combat time, and dash Mach number, while the strike mission only strongly depends on combat radius.

4.3 Thrust vs. Reference Area

The T/W-W/S plots are converted to the to thrust (T) vs. wing reference area (S) plots using algorithms in Martins. For the preliminary T-S plot, constraints are constructed about a given design thrust and reference area. This means that a design point that is feasible in the T/W-W/S plot is guaranteed to be feasible in the T-S plot. However, the plots yield information on the potential limits on thrust and wing area for the mission.

Using the T-S plots, the design point for the F/A-67 has a wing reference area of 670 ft^2 and a maximum installed thrust of 59,567 lb. The thrust is determined using a 10% reduction due to inlet losses and engine offtakes. Figures 6 and 7 show that the design point remains in the feasible region, highlighted in green. Additionally, contours of total flyaway cost for 500 units are plotted with the predicted value of 52.7 $B 2025. This is less than the total flyaway cost for the F/A-18E which is 56.6 $B 2025 for 500 units.

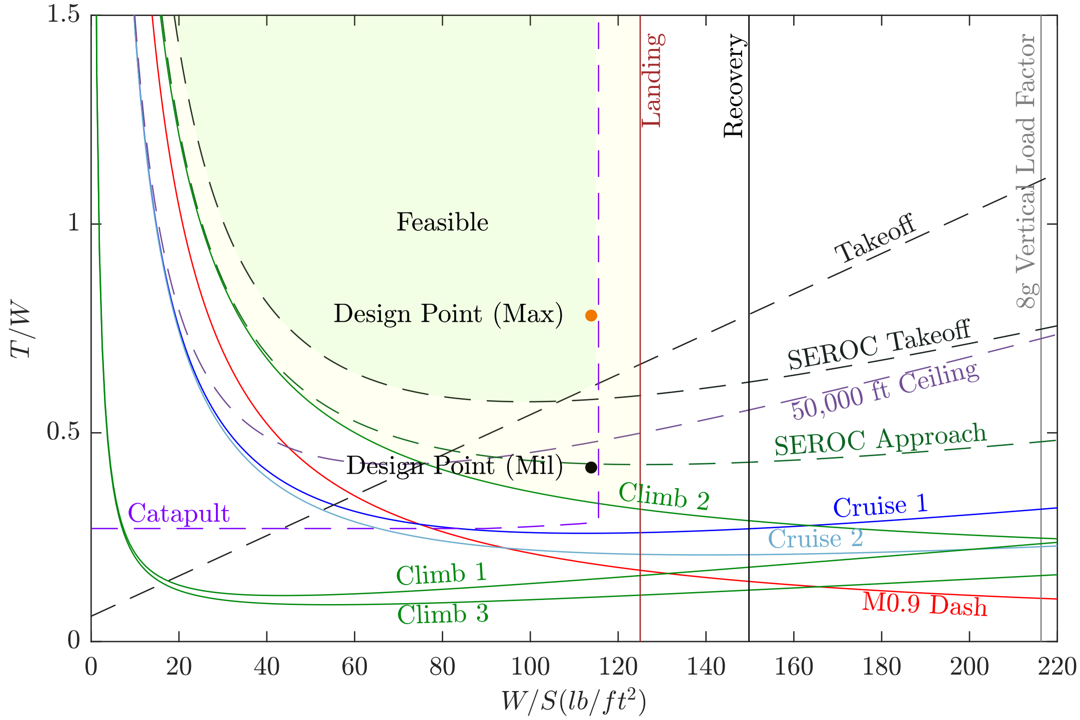

Figure 5: T/W-W/S plot for the strike mission.

4.4 Preliminary Trade Studies

Preliminary trade studies for the F/A-67 included deciding on the number of crew and engine selection. For crew, considerations were made for the benefits and drawbacks of 2 crew members over 1 crew member. Since reducing takeoff weight was a primary concern through the preliminary sizing process, a single crew member would benefit this. Furthermore, the size of aircraft, and in particular, the length, was reaching the RFP defined limit of 50 ft. Placing two crew members front and back in the cockpit would lengthen the cockpit, reducing the space available for fuel and avionics. Placing crew members side-by-side would increase the width of the fuselage, including displacing the intakes outward and potentially increasing the drag of the fuselage. However, 2 crew members would lessen the burden on each. However, the team believes that the modern avionics and equipment present in the aircraft would mitigate this issue. From a qualitative analysis, it is better to have a single crew member for reducing aircraft weight and size. This can enable the design to increase mission performance that would otherwise be sacrificed by having two crew members.

For engine selection, the preliminary trade revolved around the number of engines to use. Early analysis using sizing plots found that even for a the minimum supersonic dash Mach number of 1.6, the T/W required was high, typically > 0.5. Since increasing mission performance is a core design objective, this only increases the need for more thrust. Additionally, increasing other mission performance parameters like combat radius and combat time increased takeoff weight. Therefore, feasible designs often needed 2 engines to provide sufficient T/W. A consequence of the decision to use two engines meant the fuselage was larger, and could more easily store all ordnance internally.

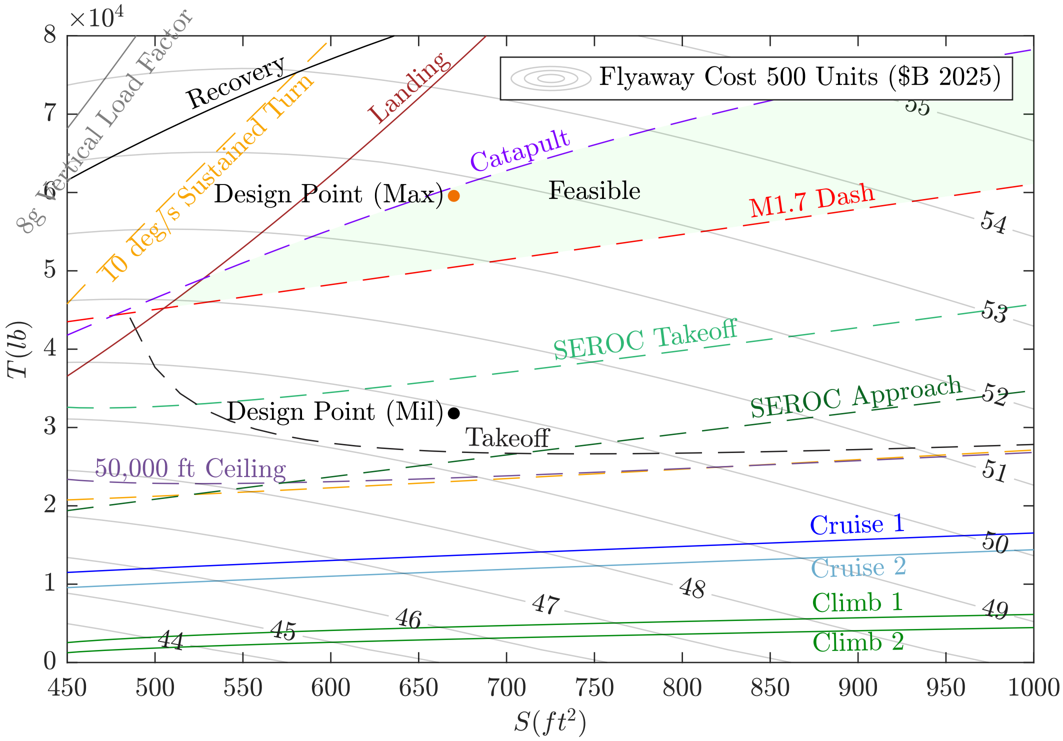

Figure 6: T-S plot for the air-to-air mission with contours of flyaway cost for 500 units at a given maximum thrust.

Therefore, selecting two engines works well with internal ordnance carriage, which is specifically mentioned to be desired in the RFP. Further details on the specific engine selection and associated justification are discussed in §9.

5 Interior Layout

5.1 Three-View Drawing

The internal layout is shown in Figure 8.

5.2 Cockpit Layout

The cockpit starts at 5.5 ft from the nose, similar to the F-35C. Internal volume of cockpit is approximately 102 ft^3. Overall cockpit dimensions are shown in Figure 9.

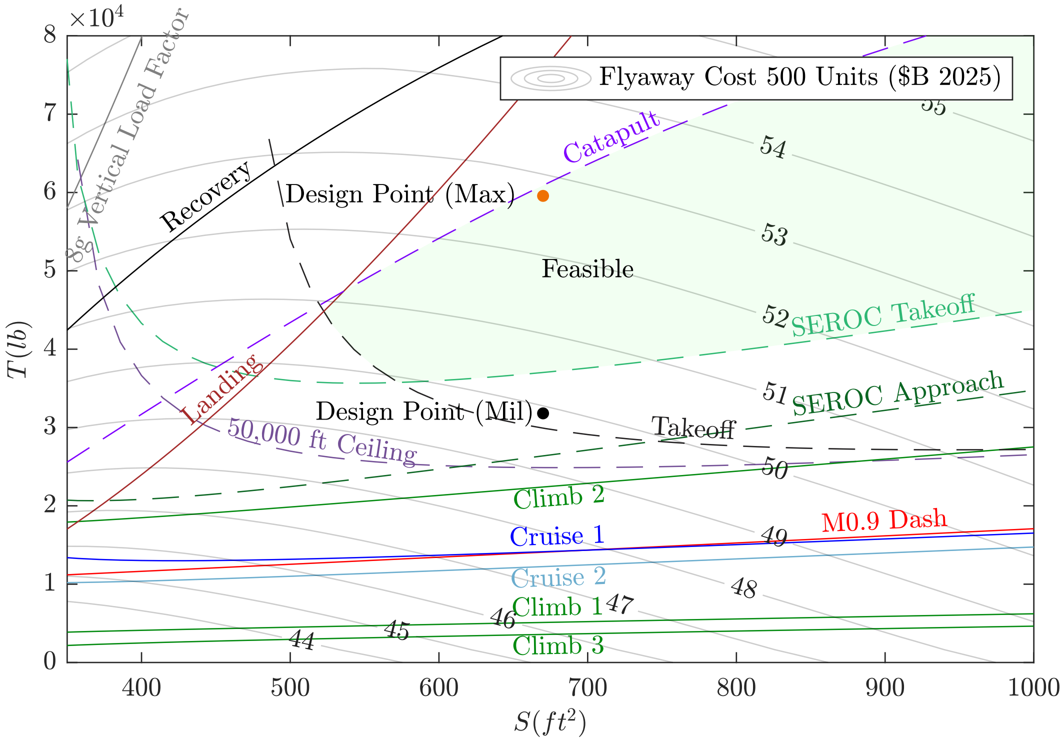

Figure 7: T-S plot for the strike mission with contours of flyaway cost for 500 units at a given maximum thrust.

5.3 Crew Selection

The aircraft is crewed by a single pilot for reduced weight and volume. One pilot is also a reasonable choice due to the systems on the aircraft, which will assist the pilot. Some of these systems are discussed in §10.1.

5.4 Pilot Sight Lines

Pilot sight lines fall within recommended angles, as shown in Figure 10. The overnose angle is 16.0 deg and the overside angle is 40.6 deg. The grazing angle is 14.1 deg.

5.5 Weapons Bay

There are two ordnance loadout configurations. The strike configuration consists of 4 MK-83 JDAMs and 2 AIM-9X Sidewinders with a total internal volume of 74 ft^3. The air-to-air configuration consists of 6 AIM-120s and 2 AIM-9X Sidewinders, leading to a required internal volume of 177 ft^3. For reduced radar cross section and drag, the weapons will be stored fully internally, as shown in Figure 11. A detailed trade study for internal vs. external storage is discussed in §14.

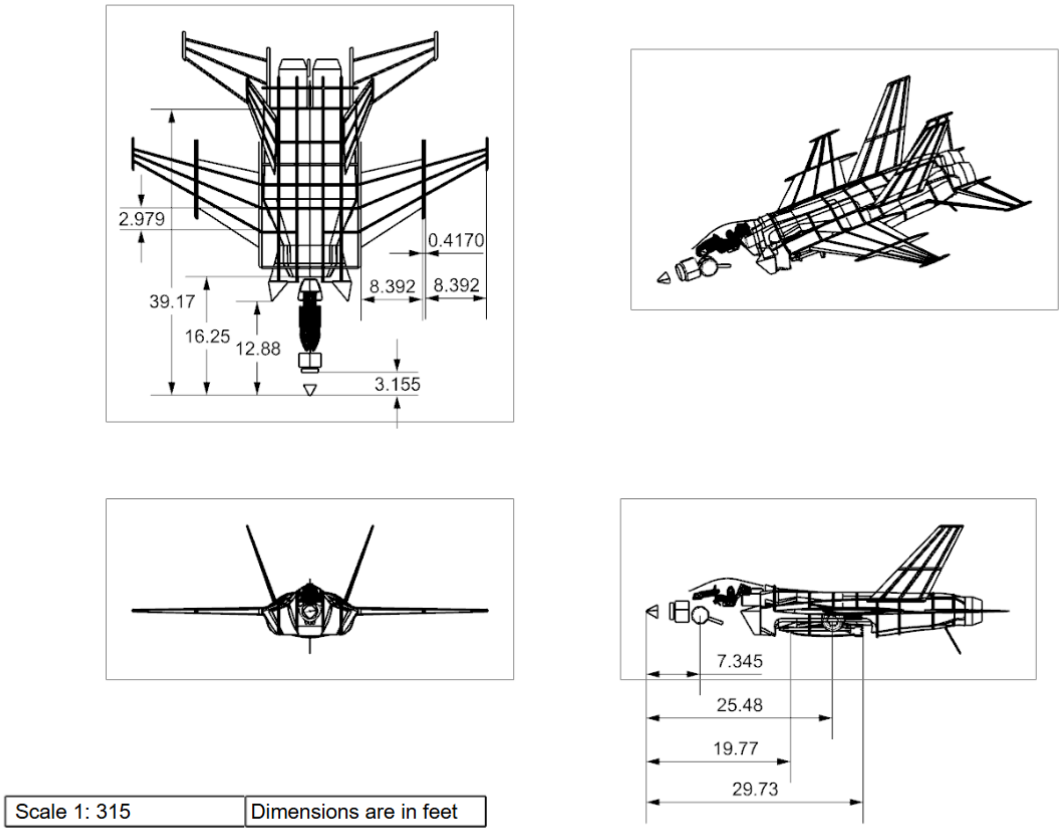

Figure 8: Interior layout three-view drawing of the F/A-67.

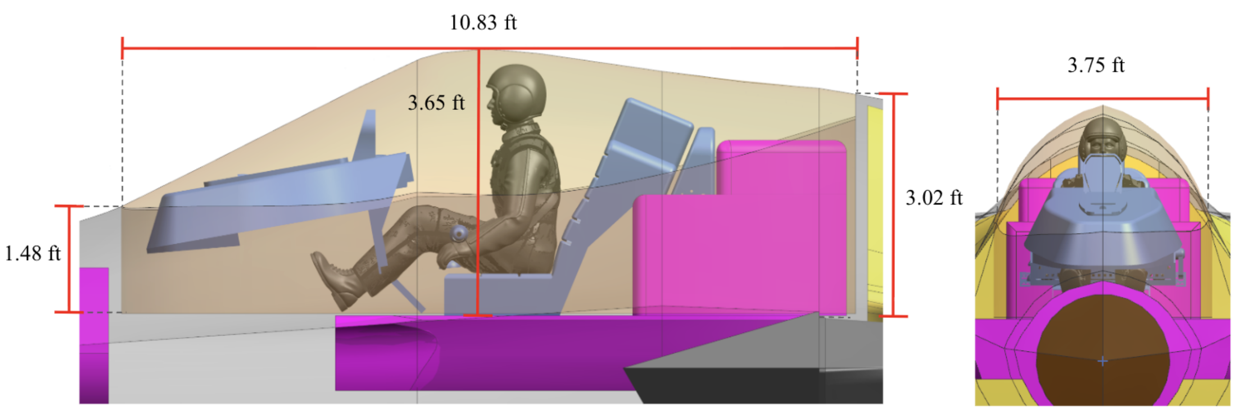

Figure 9: Dimensioned Cockpit Layout

5.6 Fuel Tanks

All fuel is located internally, due to the results discussed in §14. The majority of the fuel is within the fuselage, with some fuel in the wings and vertical tail. No fuel is stored in the control surfaces.

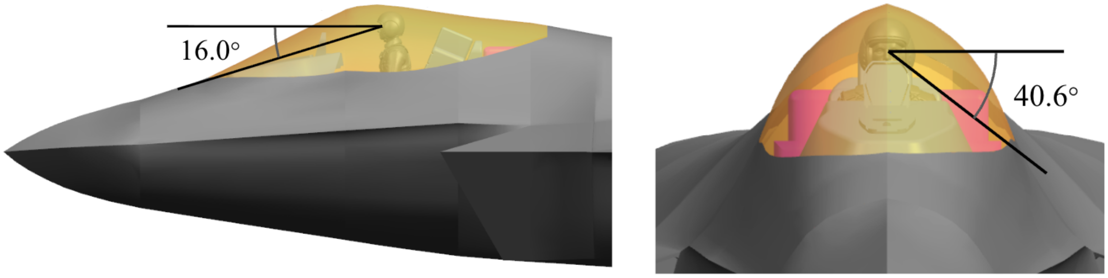

Figure 10: Overnose and overside angles.

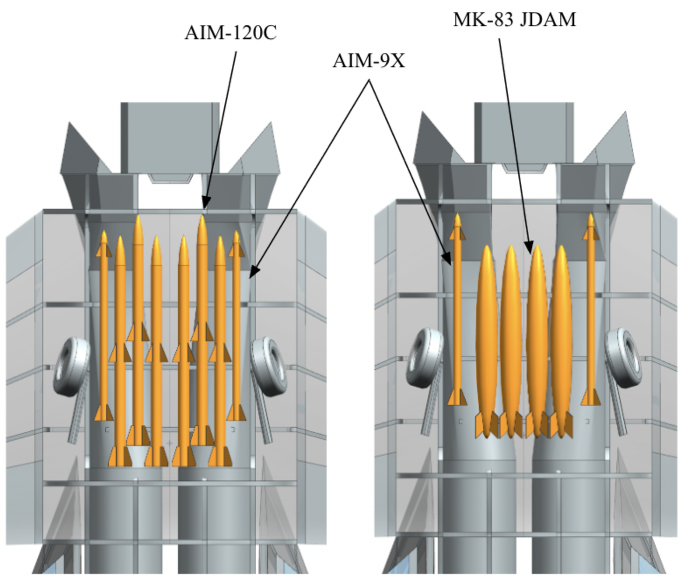

Figure 11: Weapons bay layout with ordnance shown in orange. Both configurations are stowed neatly into the fuselage within one bay.

Fuel storage within the vertical tail was deemed to be acceptable by looking at F-35C fuel tanks [7].

The volume of the fuel tanks were initially estimated by calculating the volume of a trapezoidal prism based on wing and vertical tail dimensions. This amount of fuel was subtracted from the total amount of fuel needed to get the amount of fuel in the fuselage. After further CAD refinement, the finalized volumes were modeled, resulting in a slightly higher wing fuel volume and slightly lower vertical tail fuel volume. The fuel volumes and weights are shown in Table 3 and the fuel tank locations are shown in Figure 12.

5.7 Avionics

The avionics bays are shown in Figure 13. A more detailed discussion of avionics is found in §10.

Table 3: Fuel storage locations

Fuel Location	Volume of Fuel (ft^3)	Weight of Fuel (lb)
Wings	92.2	4,648.9
Vertical Tails	22.8	1,149.6
Fuselage	419.7	21,162.1
Total	534.7	29,960.6

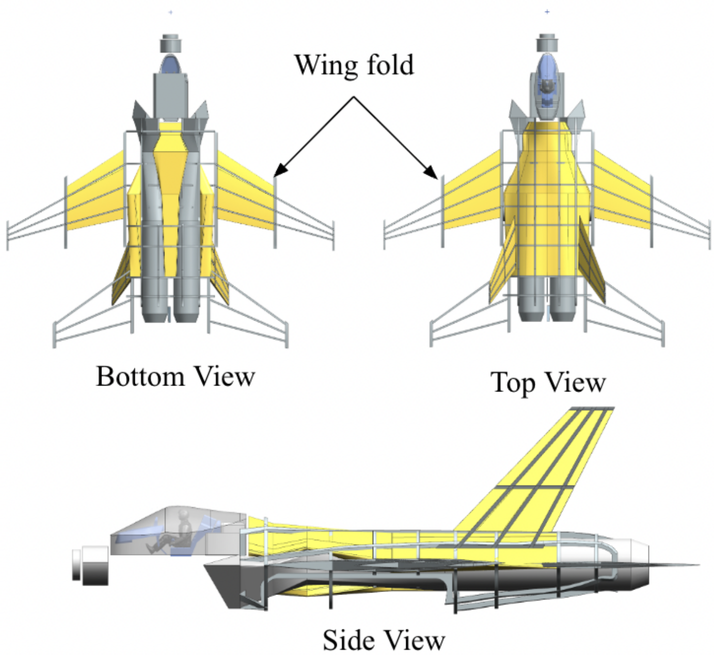

Figure 12: Fuel tanks are shown in yellow. Control surfaces are not shown.

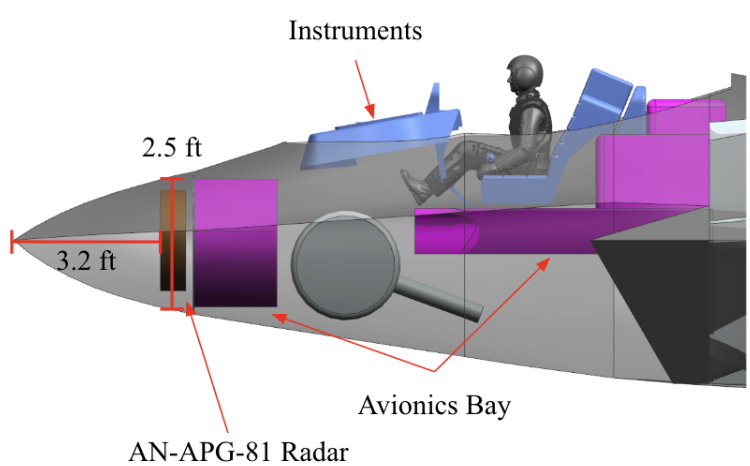

Figure 13: Internal view of avionics layout.

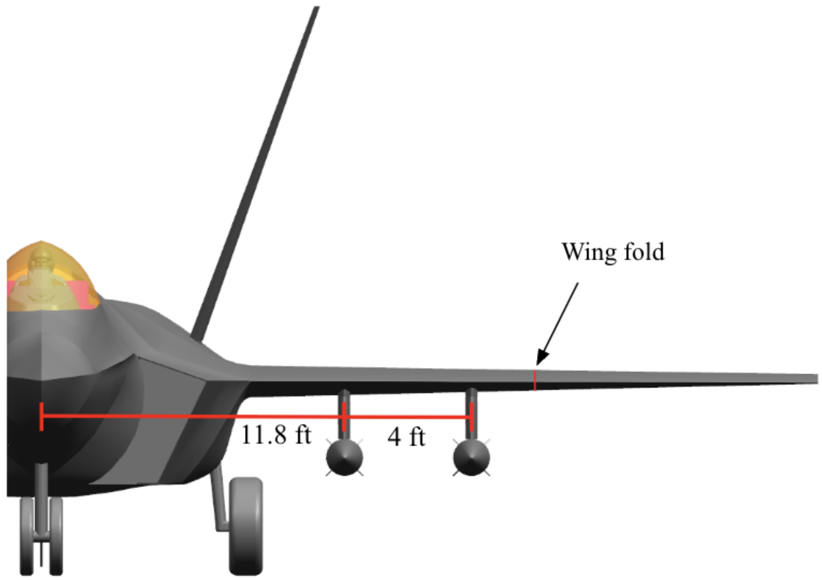

Figure 14: External removable hard point locations.

5.8 External Storage

As discussed in §14, the F/A-67 has fully internal storage for the air-to-air and strike missions. Per the RFP, the aircraft must be capable of carrying 10,000 lb of external payload, with each store capable of carrying 3,000 lb. Thus, four removable hard points are modeled as seen in Figure 14. Each hard point can carry one external fuel tank, two MK-83 JDAMs, or two AIM-120Cs, with the external fuel tank satisfying the 3,000 lb requirement at 3,235 lb.

6 Weights and Balance

6.1 Component Weight Buildup

The weight of individual components are calculated using a component weight buildup method from Raymer [5], discussed further in Appendix E. These calculations account for advanced composite materials fudge factors per component. Additionally, the components are sized for an 8g limit load factor with a 1.5 safety factor, resulting in an ultimate load factor of 12g. This results in an empty weight of 31,207 lb. The heaviest individual components of the aircraft are the wings, fuselage, and engines, as seen in Table 4.

Installed engines weights are shown in Table 5. The all-else empty category is broken down in Table 6.

Table 4: Component Weight Breakdown

Component	Weight (lb)
Wings	4,775
Fuselage	5,018
Vertical Tails	1,174
Horizontal Tail	510
Nose Gear	626
Main Landing Gear	1,045
Installed Engines	8,942
All-else Empty	9,118
Crew	200
Maximum Fuel	26,961
Air to Air Payload	2,460
Strike Payload	4,424

Table 5: Installed Engines Components

Component	Weight (lb)
Engines	7,980
Engine Mounts	150
Engine Section	87
Cooling Systems	621
Starter (Pneumatic)	103

Table 6: All-Else Component Weights

Component	Weight (lb)
Avionics	2,500
Flight Controls	1,149
Engine Controls	43
Instruments	161
Hydraulics	225
Electrical	793
Furnishings	218
Air Conditioning	355
Air Induction	908
Handling Gear	16
Fuel Systems	2,750

6.2 Center of Gravity

For the component weights given in Table 3 and 4, the CG is shown in Table 7.

6.3 CG Loading and Envelope

The CG loading diagram for air-to-air and strike is shown for two permutations of empty weight, crew, fuel, and payload in Figure 15. The two permutations cover the most fore and aft CG conditions for a given mission. The forward limit is determined by requiring less than 20% of the load on the nose gear. The aft limit is determined by requiring a greater than 25 deg longitudinal tip-over angle.

CG excursion during missions is shown using a CG and static margin (SM) envelope diagram (Fig. 16). The aft limit is set to a 0% SM. However, it is possible for the aircraft to have negative static stability due to the flight controller system. The forward limit is determined based on available elevator

Table 7: Component CG Locations

Component	x C.G. Location (ft from the nose)	y C.G. Location (ft from the centerline)	z C.G. Location (ft from the ground)
Wings	25.87	0	6.13
Fuselage	17.53	0	6.13
Vertical Tails	37.92	0	11.81
Horizontal Tail	43.40	0	6.13
Nose Gear (extended)	10.37	0	0.80
Nose Gear (stowed)	7.67	0	5.63
Main Landing Gear (extended)	29.67	0	1.82
Main Landing Gear (stowed)	26.04	0	4.41
Installed Engines	37.92	0	6.13
All-else Empty	17.93	0	6.13
Crew	11.37	0	8.43
Fuselage Fuel	26.66	0	6.68
Wing Fuel	27.18	0	6.13
Vertical Tails Fuel	37.54	0	11.81
Air to Air Payload	24.75	0	4.66
Strike Payload	24.81	0	4.66
Empty CG (gear extended)	26.10	0	6.09
Empty CG (gear stowed)	25.92	0	6.28

authority to maintain level flight with full flaps deflected. Additional discussion of SM and stability is covered in §7.

7 Stability and Control

7.1 Static Stability

The aircraft is statically stable in all three axes, as it has positive $C_{m\alpha}$, positive $C_{n\beta}$, and negative $C_{l\beta}$ for all flight conditions.

7.2 Neutral Point

The MAC of the aircraft is 15.25 ft. The subsonic neutral point is 26.62 ft from the nose, which is calculated using a refined AVL model [8]. This includes fuselage and trim effects. At supersonic speeds, the neutral point shifts backwards. This backwards shift is calculated using AeroSandbox AeroBuildup [9][10], a modified Python implementation of USAF Digital DATCOM [11]. The neutral point shift is +4.13 ft, which results in a supersonic neutral point of 30.75 ft.

The resulting SM is also dependent on the loading, which is calculated in §6.3. The highest and lowest subsonic and supersonic SM values are detailed in Table 8 and Table 9. Because the SM is near zero in some configurations, a fly-by-wire system as discussed in §10.1 is required to relieve

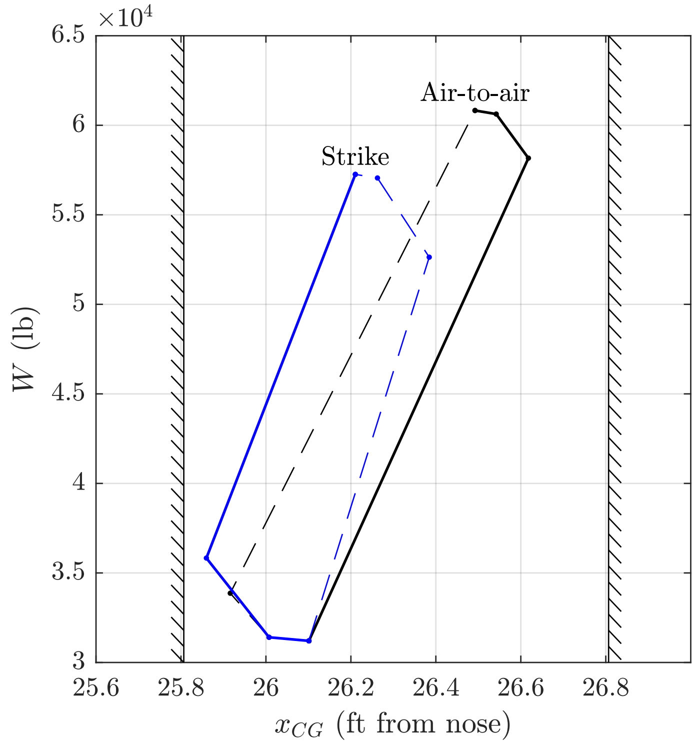

Figure 15: CG Loading diagram for Air-to-Air and Strike Missions. CG excursion is within aircraft limits.

pilot workload. The SM for the supersonic condition is large, so more investigation is needed into active fuel management to maintain supersonic maneuverability.

Table 8: Subsonic SM breakdown (%MAC).

Configuration	Air-to-Air	Strike
Most Fore	0.86	2.7
Most Aft	5.3	5.6

Table 9: Supersonic SM breakdown (%MAC).

Configuration	Air-to-Air	Strike
Most Fore	28	30
Most Aft	32	33

7.3 One Engine Inoperative Sizing

Because the aircraft has two engines, it must be capable of operating in a one-engine scenario, and the rudder must be capable of canceling the torque. The worst case for rudder control is at the takeoff condition, as the dynamic pressure is at a minimum and no sideslip angle is allowed. The torque from one engine at maximum thrust is 69,500 lbf-ft, which is equivalent to a C_n of 0.0283 at

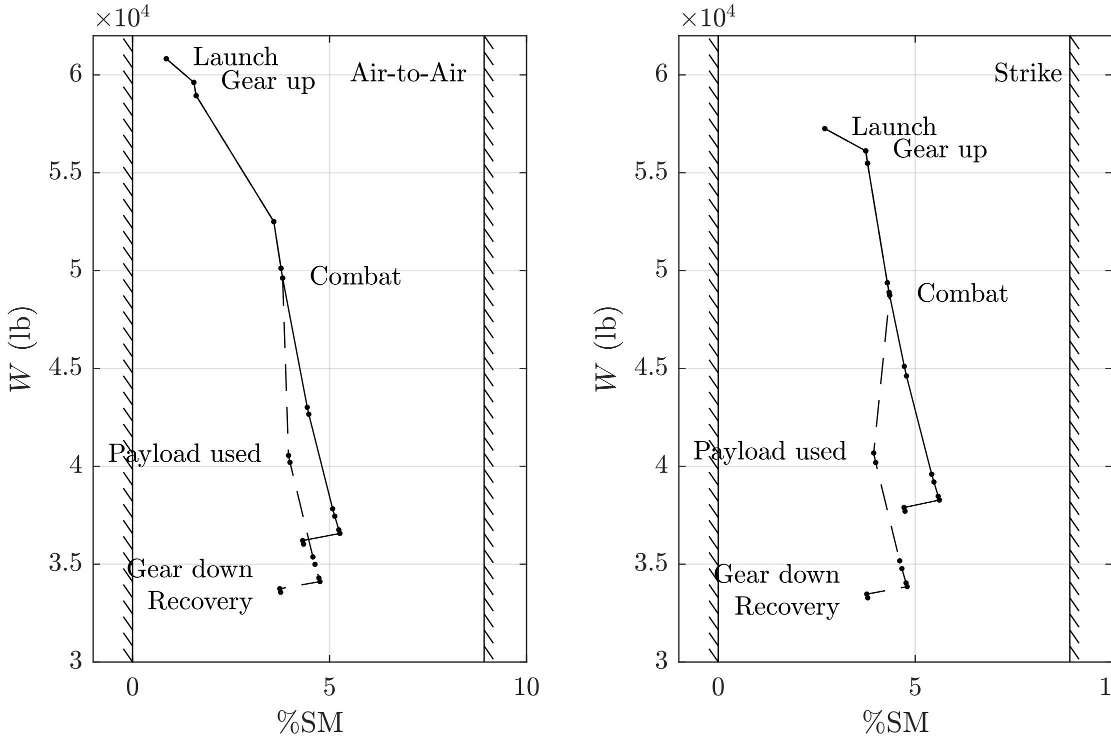

Figure 16: CG and static margin envelope diagram for Air-to-Air and Strike Missions. CG excursion is within aircraft limits.

takeoff dynamic pressure. The rudder deflection required for this C_n is 17.03 deg, which is below the Raymer [5] recommendation of 20 deg. Therefore, the rudder is properly sized for a one-engine inoperative case.

7.4 Launch and Approach Control

To ensure controllability in the launch and approach conditions, the elevator must be adequately sized to counter the flap moments, with margin for maneuverability. The worst case condition is takeoff with full flaps at minimum dynamic pressure. The corresponding elevator deflection for $C_m = 0$ is -4.25 deg, with a minimum elevator $|C_\ell|$ of 0.22. This is well below the elevator $|C_{\ell_{\max}}|$ of 0.8865, so there is margin for control at the launch and approach conditions.

8 Aerodynamics

8.1 Airfoil Design

The design and selection of the airfoils was done iteratively with the rest of the aircraft design process, as discussed in §18. The wing airfoil was designed using the MACH-Aero framework [12], and the tail airfoils are biconvex airfoils.

8.1.1 Wing Airfoil

The airfoil was designed using the MACH-Aero framework [12]. The optimizer starts with an airfoil similar to the one used by the F/A-18E, the NACA 65-206 airfoil [1]. The optimization problem is centered around two operating conditions: cruise and supersonic dash. For cruise, the design of $C_\ell = 0.55$ is calculated using the weight at the beginning of the first cruise segment. A 5-point stencil is constructed about the cruise operating condition, varying C_ℓ and Mach number. This has the effect on ensuring minimal wave drag at, and around, the design cruise condition, and not over optimizing for a single condition. For the supersonic dash condition, $C_\ell = 0.08$. The drag coefficient between cruise and dash is weighted equally.

Additional constraints include a minimum thickness ratio between the original airfoil, and a minimum volume or cross-sectional area. The thickness requirement prevents the leading and trailing edge (TE) from becoming excessively thin, while the volume constraint is useful for maintaining the fuel capacity and structural properties of the airfoil. For example, reducing area would result in a decrease in the fuel carrying capacity, while a thinner airfoil may have worse bending stiffness in a wing. The optimization problem is outlined below.

$$
\begin{aligned}
\text{minimize} \quad & 0.1C_{d,0} + 0.1C_{d,1} + 0.1C_{d,2} + 0.1C_{d,3} + 0.1C_{d,4} + 0.5C_{d,5} \\
\text{with respect to} \quad & \text{angle of attack, shape variables} \\
\text{subject to} \quad & C_{\ell,0} = 0.55, M = 0.84, h = 40000 \text{ ft} \\
& C_{\ell,1} = 0.65, M = 0.84, h = 40000 \text{ ft} \\
& C_{\ell,2} = 0.45, M = 0.84, h = 40000 \text{ ft} \\
& C_{\ell,3} = 0.55, M = 0.85, h = 40000 \text{ ft} \\
& C_{\ell,4} = 0.55, M = 0.83, h = 40000 \text{ ft} \\
& C_{\ell,5} = 0.08, M = 1.7, h = 30000 \text{ ft} \\
& V \geq V_0 \\
& t \geq 0.2847t_0
\end{aligned}
$$

The resulting airfoil has a C_d of 0.0084 at the cruise condition and a C_d of 0.0212 at the supersonic dash condition. The optimized airfoil is shown in Figure 17. The airfoil top surface has been flattened to reduce the effect of wave drag. Additionally, the leading edge (LE) is much sharper to reduce drag at the supersonic dash condition.

Plotting the c_p distribution (Fig. 18) further reveals a flat distribution on the upper surface, and a pressure increase corresponding to a weak shock. This correlates well with the description of a supercritical airfoil in Nicolai [3]. flaps will be scheduled as a function of angle of attack to reduce the LE pressure peak at off-design conditions.

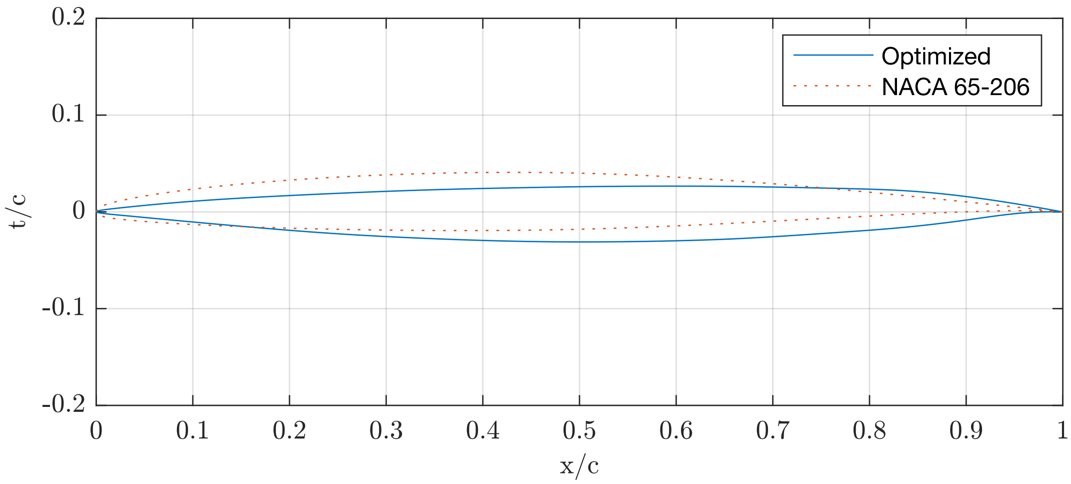

Figure 17: Optimized airfoil and initial NACA 65-206 airfoil.

8.1.2 Stabilizer Airfoil

For the horizontal and vertical stabilizer, a biconvex airfoil with a t/c of 4% was selected, based on example sharp-nosed airfoils in Nicolai [3]. This provides a symmetric shape for the stabilizers, and has increased thickness along most of the chord when compared to a double wedge airfoil. The increased thickness can be beneficial for reducing structural weight, and increase internal volume for storing fuel in the vertical stabilizers. Furthermore, based on B factors provided in Nicolai, biconvex airfoils have a lower B-factor when compared to a hexagonal airfoil. This reduces wave drag that will occur during supersonic flight.

8.2 Wing and Tail Design

The AVL [8] model of the aircraft was updated from the preliminary design to include the fuselage and tails, as well as the elevator, rudder, flaps, and flaperons. The `COMPONENT` keyword was used to join the lift distributions of the wing and fuselage.

8.2.1 Wing Layout

The design variables changed were the wing taper ratio (λ), wing quarter-chord sweep angle ($\Lambda_{0.25}$), wing twist angle (θ_{twist}), wing x-location from the nose (x_{wing}), and horizontal stabilizer moment arm length (ℓ_{HT}). The average cruise C_L of 0.4323 was used. The objective of the study was to minimize total cruise drag.

Because of the large number of design variables, some assumptions were made to simplify the trade process. First, root incidence was set to zero, which is typical for fighter aircraft according to

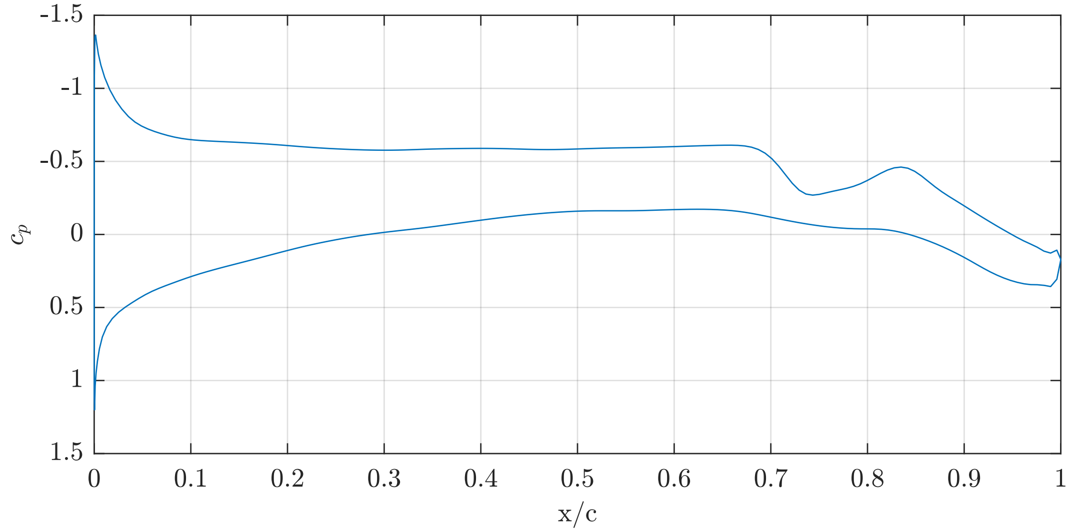

Figure 18: Optimized airfoil c_p distribution at the cruise operating condition.

Raymer [5]. Second, the CG location was assumed to be fixed in order to determine an x_{wing} that gives near neutral stability. Lastly, the optimal $\Lambda_{0.25}$ was assumed to be 30 deg, the lowest which results in zero wave drag, in order to increase $C_{L,\max}$.

For modeling of the 3D aerodynamic characteristics, AVL [8] was employed. The drag components considered were the induced drag $C_{Di,\text{wing}}$, wing parasitic drag $C_{Dp,\text{wing}}$, and tail parasitic drag $C_{Di,\text{tail}}$. $C_{Di,\text{wing}}$ was computed in AVL, and read from the output `CDff`. $C_{Dp,\text{wing}}$ was factored in using the drag polar of the wing airfoil. This was input to AVL using the `CDCL` keyword, and read through the output `CDvisc`. The `CDCL` points were tailored to achieve greatest accuracy where C_ℓ is between 0 and 0.6, matching the wing C_ℓ distribution. The $C_{Di,\text{tail}}$ was calculated using the Raymer [5] process discussed in Appendix G.

An iterative process was used to find the optimal θ_{twist} for each λ. There are three key constraints: the SM is near zero, the wing and horizontal tail do not overlap, and the horizontal and vertical tail sweeps are about 5 deg greater than the wing. The horizontal tail was placed as close to the wing as possible to meet the 50 ft length constraint.

The design was iterated upon manually using an Excel tool that automatically interfaces with AVL using Visual Basic until the C_D was minimized. The resulting design has a λ of 0.2, a θ_{twist} of -2.7 deg, an x_{wing} of 12.58 ft from the nose, and a ℓ_{HT} of 18.5 ft. The wing Λ_{LE} is 37.19 deg, the horizontal tail Λ_{LE} is 42.18 deg, and the vertical tail Λ_{LE} is 42.20 deg. This satisfies the 5 deg requirement.

8.2.2 Tail Layout

The horizontal tail AR was selected to be 3, and the λ was selected to be 0.3 based on recommendations from Raymer [5]. The tail volume coefficient (c_{HT}) was chosen to be 0.4, and was reduced by 10% because the elevator is all moving, resulting in a c_{HT} of 0.36. The moment arm length is 18.5 ft to meet the RFP 50 ft length constraint.

The vertical tail AR was selected to be 1.4, and the λ was selected to be 0.4. This is to reduce the root chord while fitting in Raymer [5] recommendations. The moment arm length was selected to be 13.2 ft, which was positioned so that the TE of the vertical stabilizer root was aligned with 50% of the horizontal stabilizer root. This ensures the rudder and elevator do not hit each other when maximally deflected.

8.3 Control Surface Sizing

The elevator is all-moving, which reduces the horizontal tail size and increases maneuverability. The average chord fraction of the rudder was determined to be 30% based on Raymer [5]. The full span of the tail is used to maximize maneuverability.

The ailerons were sized primarily based on Raymer [5]. A chord fraction of 25% was selected to match the inboard TE flap, which results in a span fraction between 64% and 100%. The folding location was placed here to ensure manufacturability.

For the LE flap, an average chord fraction of 15% along the usable span was selected using Raymer [5] guidelines. For the TE flaps, a chord fraction of 25% was selected. The usable span inboard of the folding location was used, resulting in a span fraction of 27% to 64%.

8.4 Drag Breakdown and Polars

The drag of the aircraft is estimated using a component buildup method from Raymer [5]. The equations used are given in Appendix G for subsonic parasite drag, supersonic parasite drag, wave drag, drag due to lift, flap drag, and trim drag.

Parasitic drag is estimated for three flight conditions: cruise, takeoff, and approach. During cruise, flaps, landing gear, and the arresting hook are not deployed. For takeoff, flaps are full and landing gear are deployed. For approach, flaps are half and landing gear and the arresting hook are deployed. Landing gear and the arresting hook are considered as miscellaneous drag. Flap drag is a fixed increase in C_{D0}. Leaks and protuberances is not considered to be influenced by flap drag. Wave drag is only modeled for the clean aircraft state (no flaps, landing gear stowed), and should only be considered for the cruise condition.

Figure 19 displays the breakdown of parasitic drag. Half and full flaps cause a large increase in C_{D0}, especially for full flaps. The parasitic drag coefficient nearly doubles compared to the drag at cruise. For the cruise condition, wave drag is the largest contribution to parasitic drag at supersonic speeds.

Drag polars were also generated for the same three flight conditions at their nominal Mach numbers (Fig. 20). This is Mach 0.84 for cruise, and the respective speeds for takeoff and approach. The drag polars show the clear increase in C_{D0} due to flaps and landing gear. The increase in $C_{L,\max}$ due to flaps is also shown. Trim drag is modeled for all conditions as a 5% increase to K.

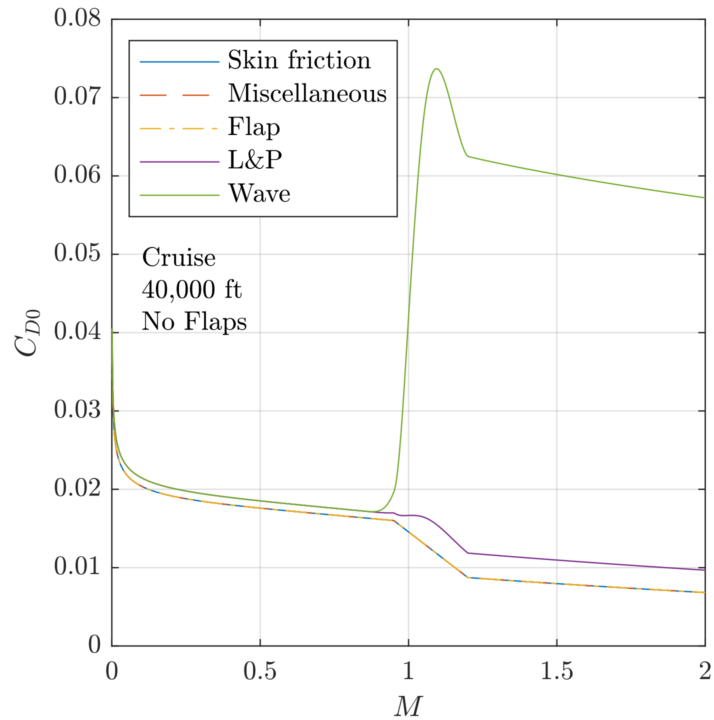

Figure 19: Parasitic drag breakdown for cruise operating condition.

8.5 $C_{L_{max}}$ Estimation

This section discusses the calculation of the aircraft's $C_{L_{max}}$ for the clean, approach, and takeoff conditions.

8.5.1 2D $C_{\ell_{\max}}$

The aircraft will use plain LE flaps and slotted TE flaps to increase the $C_{L_{\max}}$. Because the aircraft uses slotted flaps, it is not possible to model the $C_{\ell_{\max}}$ using XFOIL. A multi-element code such as MSES is required, which is unavailable to the team. Therefore, lower fidelity estimation methods are used.

First, the clean $C_{\ell_{\max}}$ must be found. Because of the wing airfoil's sharp LE and high Reynolds number, it cannot be analyzed by XFOIL or ADflow at high angles of attack. Instead, experimental data can be used to approximate the $C_{\ell_{\max}}$. In NACA TN2502 [13], the stall behavior of the NACA 64A006 airfoil is studied at a Reynolds number of 5.8×10^6. This is assumed to behave similarly to the wing airfoil, as it exhibits LE stall. The Reynolds number is also comparable to the takeoff Reynolds number of 17×10^6. The $C_{\ell_{\max}}$ of the NACA 64A006 is 0.8865, taken from data shown in Appendix H.

To find the flapped $C_{\ell_{\max}}$, empirical adjustments from Raymer are used. These are in the form of

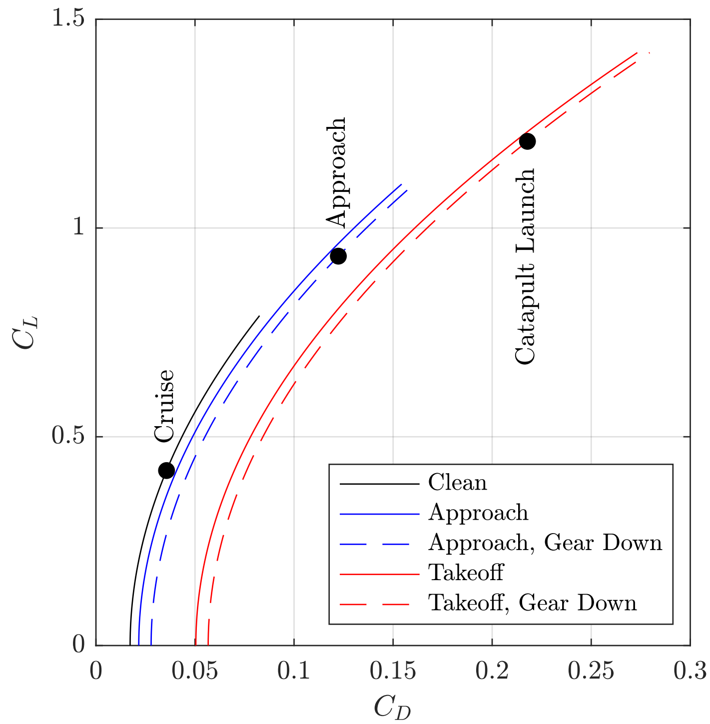

Figure 20: Drag polars for cruise, takeoff, and approach flight conditions. The cruise point at mid-mission weight is marked. Curves terminate at $C_{L,\max}$.

$\Delta C_{\ell_{\max}}$ values, which is 0.9 for a plain flap, 1.3 for a slotted flap, and 0.3 for a LE flap. There are two different airfoil sections along the wing, listed in Table 10. The associated total $\Delta C_{\ell_{\max}}$ and $C_{\ell_{\max}}$ values are also listed. Because the Raymer empirical $\Delta C_{\ell_{\max}}$ does not factor in flap deflection, they are assumed to be at full flaps. The $\Delta C_{\ell_{\max}}$ for half flaps (approach) is therefore assumed to be half of the full flapped values.

Table 10: Airfoil Sections

Section	LE Device	TE Device	Full $\Delta C_{\ell_{\max}}$	Full $C_{\ell_{\max}}$	Half $\Delta C_{\ell_{\max}}$	Half $C_{\ell_{\max}}$
1	None	None	0	0.8865	0	0.8865
2	Plain	Slotted	1.6	2.4865	0.8	1.6865

8.5.2 3D $C_{L_{\max}}$

The $C_{L_{\max}}$ of the aircraft for each configuration is estimated using AVL using the critical section method. The C_L is increased by increasing the angle of attack until the wing C_ℓ exceeds the corresponding $C_{\ell_{\max}}$ value in Table 10. The tail is trimmed to $C_m = 0$. The corresponding $C_{L_{\max}}$ and V_{stall} values are listed in Table 11. Note that the segment weight also changes the V_{stall}.

The predicted $C_{L,\max}$ is lower than the values for preliminary design, because the taper ratio decreased and the twist angle stayed the same, resulting in a stronger tip stall. Further investigation

Table 11: Configuration $C_{L_{\max}}$ and V_{stall}

Condition	Configuration	$C_{L_{\max}}$	V_{stall} (kts)
Cruise	Clean	0.79	351
Approach	Half Flaps	1.11	131
Takeoff	Full Flaps	1.42	144

is required to increase the $C_{L_{\max}}$, which could result in a smaller wing and a lighter aircraft.

9 Propulsion System

The aircraft is equipped with a dual-engine configuration. The selected engine is the F110-GE-132, providing a maximum uninstalled thrust of 33,093 lb per engine and an intermediate thrust of 17,669 lb per engine. This engine meets the RFP requirement of an existing production design as it is currently used on the F-16E/F [14].

9.1 Engine Location and Inlet Ducts

The F110-GE-132 has a maximum diameter of 3.88 ft, an engine inlet diameter of 2.97 ft, a total length of 15.15 ft, and are located 2.33 ft from the aircraft centerline to the center of the engines. The engine inlets are supersonic diverterless inlets, configured with forward-swept cowls, which divert the boundary layer away from the inlet and reduce RCS. Each duct inlet area is 7.245 ft^2.

9.2 Engine Selection

Out of the engines considered, the F110-GE-132 has the highest maximum thrust with lower dry thrust specific fuel consumption (TSFC). This engine meets the required thrust for the missions after accounting for offtakes and installed thrust corrections.

Table 12: Engine comparison table

Engine	Thrust Wet (SLS-lb)	Intermediate Thrust (SLS-lb)	Weight (lb)	TSFC (Dry) (1/s)	TSFC (Wet) (1/s)
F110-GE-132	33,093	17,669	3,990	0.68	1.90
F414-GE-400	21,496	14,447	2,512	0.82	1.844
F110-GE-129	29,474	17,595	3,940	0.67	1.85
F110-PW-229	29,100	17,800	3,850	0.74	2.05
F110-PW-229A	32,500	20,100	3,830	0.71	1.86

9.3 Thrust Offtakes

Engine performance takes into account atmospheric thrust lapse, ram effects, and power and airflow offtakes. Power and airflow offtake systems can be accounted for through thrust corrections and installed engine corrections. The process for determining these is discussed in Appendix C.

10 Government Furnished Equipment

The F/A-67 avionics will be placed closer to the nose of the aircraft to reduce the length of the wires and complexity of the system. The avionics, sensors, and systems outlined in this section will be the major components included in the 2,500 lb required by the RFP. The avionics bay is shown in Figure 13.

10.1 Avionics and Sensors

The F/A-67 will feature the AN/APG-81 actively electronically scanned array (AESA) radar [15], which will be incorporated into the nose of the aircraft. This is seen in Figure 13. The radar was chosen because it is the most modern fighter radar offered by the military with a TRL of 9. The AN/APG-81 AESA radar is newer than the AN/APG-79 [16], which is used on the F/A-18E/F. It is currently used on the F-35 to detect and differentiate between enemy aircraft and friendly aircraft, and to guide ordnance in all weather conditions. The AN/APG-85 was also considered, since it is the most modern radar in development by Northrop Grumman [17]. However, the TRL level could not be confirmed to be 6 or higher due to the many delays associated with its fitting on the F-35 [18].

Additionally, the joint precision approach and landing system (JPALS) will be integrated onto the F/A-67 [19]. The JPALS is the more modern equivalent of the automatic carrier landing system (ACLS) used on older F/A-18E/F models. JPALS is crucial for the F/A-67 since it enables the aircraft to approach and land in all weather conditions on aircraft carriers.

The F/A-67 will also incorporate a quadruple redundant fly-by-wire (FBW) system as on the F-A/18 according to Jane's [20]. This is to ensure stability for relaxed SM configurations. The quadruple redundancy guarantees that the FBW system will remain operational throughout the mission.

The AN/ALQ-214 Integrated Defensive Electronic Countermeasures (IDECM) was selected as a replacement for the AN/ALQ-126B and the AN/ALQ-165 used in the PDR report [21]. The Advanced Electronic Warfare (ADVEW) system now replaces the AN/ALQ-214 IDECM and the AN/ALR-67 system to save cost and space [22]. ADVEW allows the F/A-67 to use jamming and decoy countermeasures to confuse enemies. Additionally, it can warn against hostile radar use. The ADVEW system is TRL 6 according to [23]. In terms of electronic warfare, the AN/ALQ-214 IDECM system will be utilized [21].

Finally, the last major avionics system the F/A-67 will feature is an Advanced Mission Computer. This system is modular, allowing it to implement new software, and it replaces the old AN/AYK-14 digital computer on the F/A-18E/F for weapon systems [24].

10.2 Life Support

The F/A-67 integrates the Genesis III helmet into the cockpit [25][26]. The Genesis III provides flight data, night vision, and 360° exterior view. Additionally, the On-Board Oxygen Generation System (OBOGS) is integrated into the cockpit [27]. The OBOGS allows for the pilots to sustain high altitudes and high-g maneuvers. Finally, the cockpit features a US16E ejection seat from the F-35 [28]. The US16E meets the Neck Injury Criteria defined by the US Government, ensuring safe ejection of the pilot.

10.3 Ordnance

As discussed in § 14, the ordnance will be stored internally for reduced drag and RCS. The air-to-air mission ordnance consists of six AIM-120Cs and two AIM-9Xs. The strike mission consists of four MK-83 JDAMs and two AIM-9Xs. The internal weapons bay is shown in Figure 11.

11 Landing Gear and Carrier Suitability

The F/A-67 features a tricycle landing gear configuration. The main landing gear has one tire, while the nose gear has twin wheels in order for catapult shuttle attachment. This landing gear configuration was selected as it is seen in many carrier-based fighters, such as the F/A-18E/F [29].

11.1 Tire Sizing

Main landing gear tire size was calculated using equations 8 and 9 from [5], which take in the static weight on the wheels and output the tire diameter and width. These formulas were used to size the main tires.

$$D_{\text{main}}\,(\text{in}) = 1.59\,W_W^{0.302} \tag{8}$$

$$W_{\text{main}}\,(\text{in}) = 0.0980\,W_W^{0.467} \tag{9}$$

Additional safety factors of 7% and 25% are added to the weight on wheel, based on recommendations in Raymer [5], and to account for future mass growth. This results in a main tire diameter of 3.12 ft, and a main tire width of 1.08 ft.

When using equations 8 and 9 for the nose tires, the diameter and width are smaller than historically recommended due to lower nose wheel loads. Raymer [5] states that the nose tire can be sized to between 60-100% of main wheel dimensions. Therefore, it is assumed that nose tire dimensions are 80% of main tire dimensions.

This results in a nose tire diameter of 2.49 ft and a width of 0.86 ft. Because there are two nose tires, each tire width is 0.43 ft.

Tire size was also calculated using equation 10, which calculates the kinetic energy dissipated on landing. However, this was not as constraining as the calculation using equations 8 and 9. These equations are valid for our aircraft, as it must conduct landings on non-carrier runways.

$$KE_{\text{braking}} = \frac{1}{2}\frac{W_{\text{landing}}}{g}V_{\text{stall}}^2 \tag{10}$$

11.2 Landing Gear Placement

The ratio of distances of the main and nose landing gear from the center of gravity was calculated using a moment balance. The distances were selected so that more than 80% of the static load is transmitted to the main landing gear at any CG case.

The distance between the main gear and nose gear was manually selected to satisfy tipback and longitudinal tipover constraints. Furthermore, the width of the main gear was selected to satisfy the overturn angle constraint. These angle constraints come from Raymer. Table 13 summarizes the F/A-67 landing gear angles and historical recommendations.

The wingstrike angle was determined to ensure there is no risk of wing strike on takeoff and landing. This constraint is given by Roskam [4]. This shows that overturn is more constraining than wingstrike. As seen in Table 13, all landing gear angle constraints are satisfied.

Table 13: Landing Gear Angles

	Angle (deg)	Recommendation (deg)
Tipback (Tailstrike)	14.27	10-25
Longitudinal Tip-over (Main wheel to CG)	26.42	>Tip-back, >25
Overturn	50.09	<54
Wingstrike	18.54	>5

11.3 Strut Height

The stroke length is calculated using equation 11 from [5].

$$l_{\text{stroke}} = \frac{V_{\text{vertical}}^2}{2g\eta N_{\text{gear}}} - \frac{\eta_T}{\eta}S_T \tag{11}$$

The sink rate, V_{vertical}, is 22 ft/s from [4]. N_{gear} is the gear load factor, taken from [5]. η is the shock absorber efficiency, and η_{n} is the tire shock absorber efficiency. S_T is the tire stroke length.

The resulting stroke length is 1.24 ft. Raymer lists that the oleo total length is approximately 2.5 times the stroke length. Therefore, the minimum strut length is 3.12 ft. The minimum length was selected in order to satisfy the RFP height constraint. Similarly, for the nose gear a stroke length of 1.30 ft was found, resulting in a 3.26 ft minimum strut length.

11.4 Stowed Gear

The landing gear are stowed to reduce drag in flight, as Roskam recommends stowing gear if the aircraft cruise speed is above 150 kts [4]. The main gear folds forward and is stowed inside the fuselage. The nose gear folds forward and up into the nose.

11.5 Landing Gear Retraction and Arrestment Hook

The landing gear for stowed and extended configurations, as well as the arrestment hook, are shown in Figure 21.

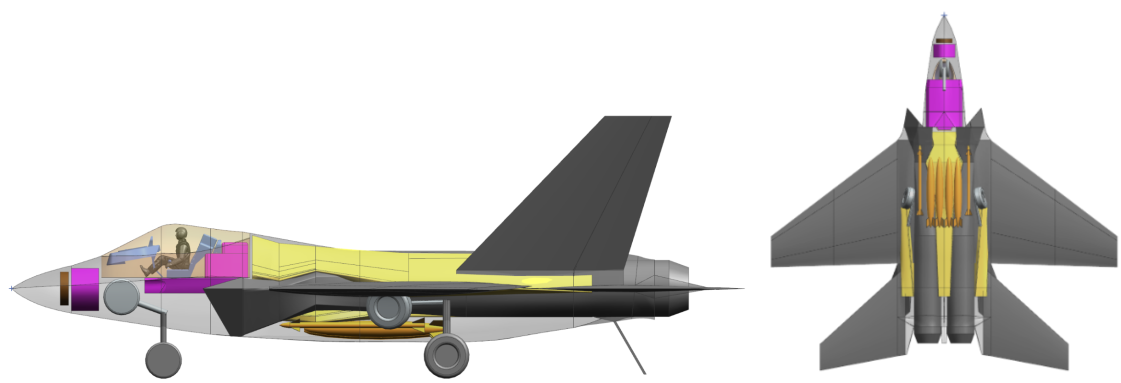

Figure 21: Side (left) and bottom (right) view of landing gear extended and stowed. Arrestment hook is shown in the side view.

11.6 Launch Performance

Based on the F/A-67 takeoff weight, the catapult endspeed can be calculated using catapult performance charts provided in the RFP (Fig. 22). The performance curves are used during the sizing process for constraint curves. The deadload weight is assumed to be the takeoff weight of 60,826 lb. The corresponding catapult endspeed and peak acceleration is 156 kts and 4g respectively.

The F/A-67 also satisfied all four points outlined by the RFP for catapult performance. All requirements are evaluated for tropical day conditions at 89.8°F with zero WOD.

1. The aircraft's high T/W gives the F/A-67 a high takeoff SEROC of 3198 ft/min evaluated at the catapult endspeed. This exceeds the 200 ft/min minimum so it is reasoned that this satisfies the RFP 10 ft sink constraint.

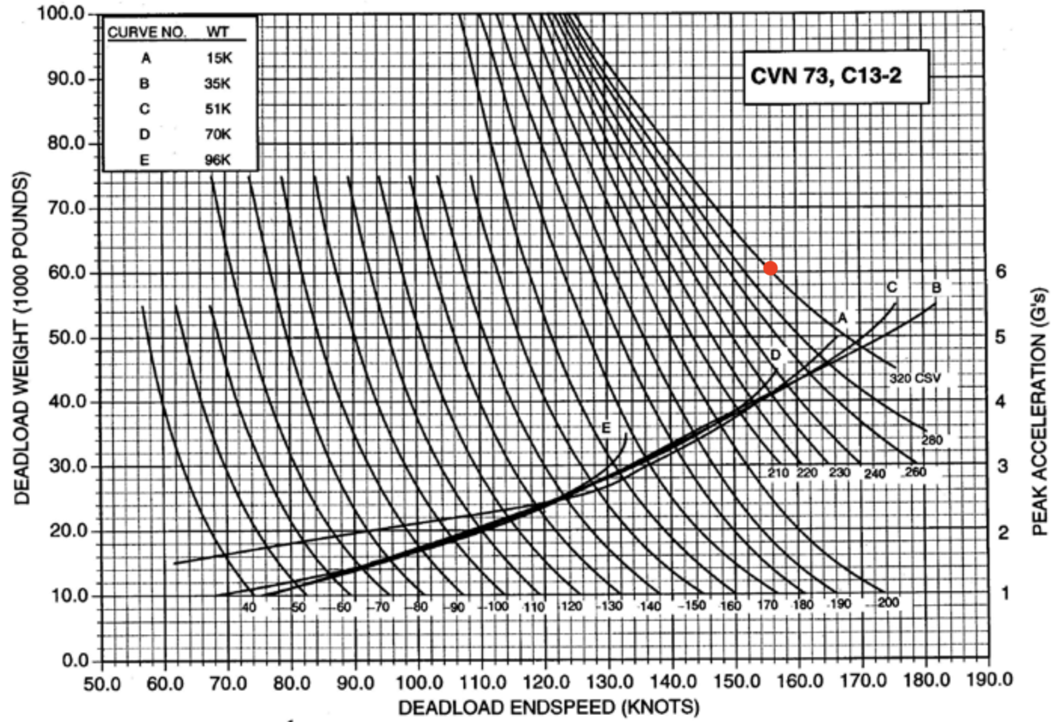

Figure 4-14

TYPICAL C13-2 MINIMUM PERFORMANCE AND LOAD FACTOR CURVES

Figure 22: C13-2 Catapult performance curves with the F/A-67 takeoff weight and catapult endspeed marked.

2. The aircraft takeoff $C_L = 1.21$ is less than $0.9C_{L,\max} = 1.28$.
3. The aircraft greatly exceeds the required T/W necessary for > 0.065g longitudinal acceleration. This is enforced in the sizing plots shown in Figures 27 and 28
4. One engine inoperative controllability is satisfied through the rudder sizing. 17.03 deg of deflection is needed to counteract the moment generated by a single engine at maximum thrust, which is less than the 20 deg recommended limit [5].

11.7 Recovery Performance

Recovery performance is evaluated at the maximum landing weight specified in the RFP. Figure 23 shows that for the landing weight of 39,177 lb, the engaging speed of 134 kts does not exceed the limits of the arresting gear. This result assumed of 15 kts WOD which reduces the aircraft stall speed to 116 kts. The corresponding approach speed is then 127 kts which is less than the 145 kts limit in the RFP. During arrestment, the F/A-67 has a hookload of 145,000 lb. During approach, the aircraft SEROC is 5253 ft/min which exceeds the 500 ft/min minimum.

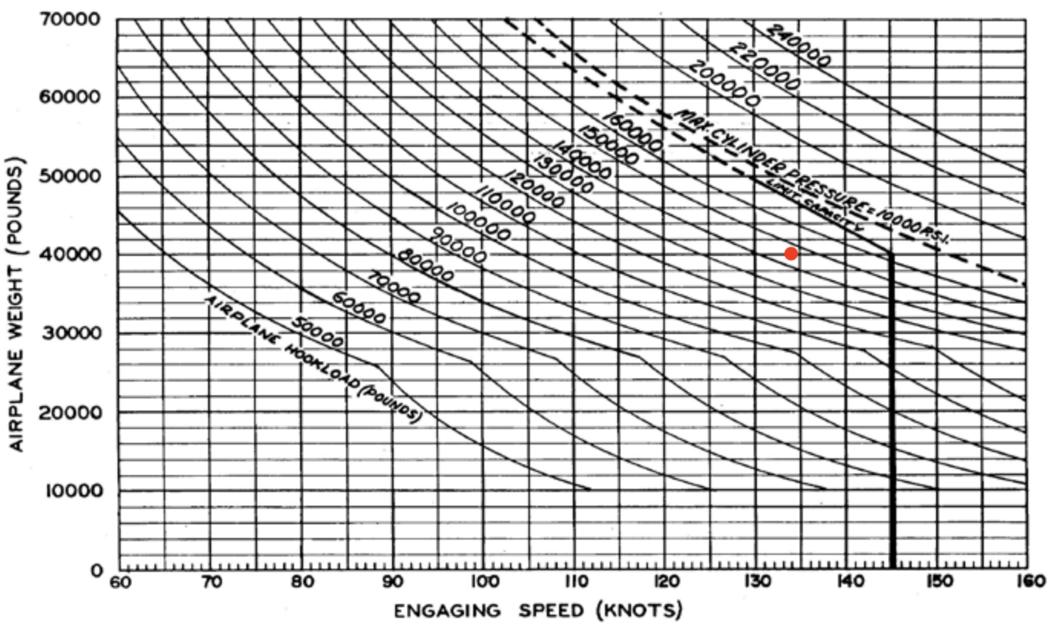

Figure 23: Arresting gear performance curves with the engaging speed of the F/A-67 marked.

12 Method Validation

The takeoff weight and sizing plot codes were validated against the weights and mission performance parameters of the F/A-18E. Mission profiles were selected to be similar to the air-to-air and strike missions outlined by the RFP. Aircraft parameters were modified to match the wing area, AR, payload weight, and engine performance of the F/A-18E. Mission performance parameters like combat radius were set using values derived from [1]. Geometric parameters used for more detailed weight estimation and drag buildup were neglected.

12.1 Weight Estimation

The predicted empty and takeoff weights are compared against the actual values in Tables 15 and 14, and are shown to be reasonably close to the actual weights. The empty weight is underestimated by roughly 2,000 lb. This may be due to various composite weight correction factors used in the component empty weight analysis. The takeoff weight for the air-to-air mission is similarly close to actual value, but the strike mission weight is off by a larger margin of 10,000 lb. This may be due to the refined fuel fraction estimation methods, since they consider optimal climbs and cruises. The actual mission profiles for the F/A-18E may differ, producing differing weights. However, since the empty weight estimate is reasonable, and since fuel fraction depends on many aircraft specific parameters that were not determined, the team is confident in the validity of the preliminary weight estimation for the F/A-67.

Table 14: Empty weight validation for the F/A-18E.

Mission	Estimated Empty Weight (lb)	Actual Empty Weight (lb)
Air-to-Air	28,891	30,554
Strike	28,891	30,554

Table 15: Takeoff weight validation for the F/A-18E.

Mission	Estimated Takeoff Weight (lb)	Actual Takeoff Weight (lb)
Air-to-Air	46,702	48,247
Strike	50,158	60,769

12.2 Refined Sizing

The results of the weight estimation procedure are input into the sizing plot codes. The produced T/W-W/S plots indicate that the F/A-18E is feasible for both missions. This suggests that the sizing codes are valid, since they predict the F/A-18E is feasible. The plots are shown in Figures 24-25.

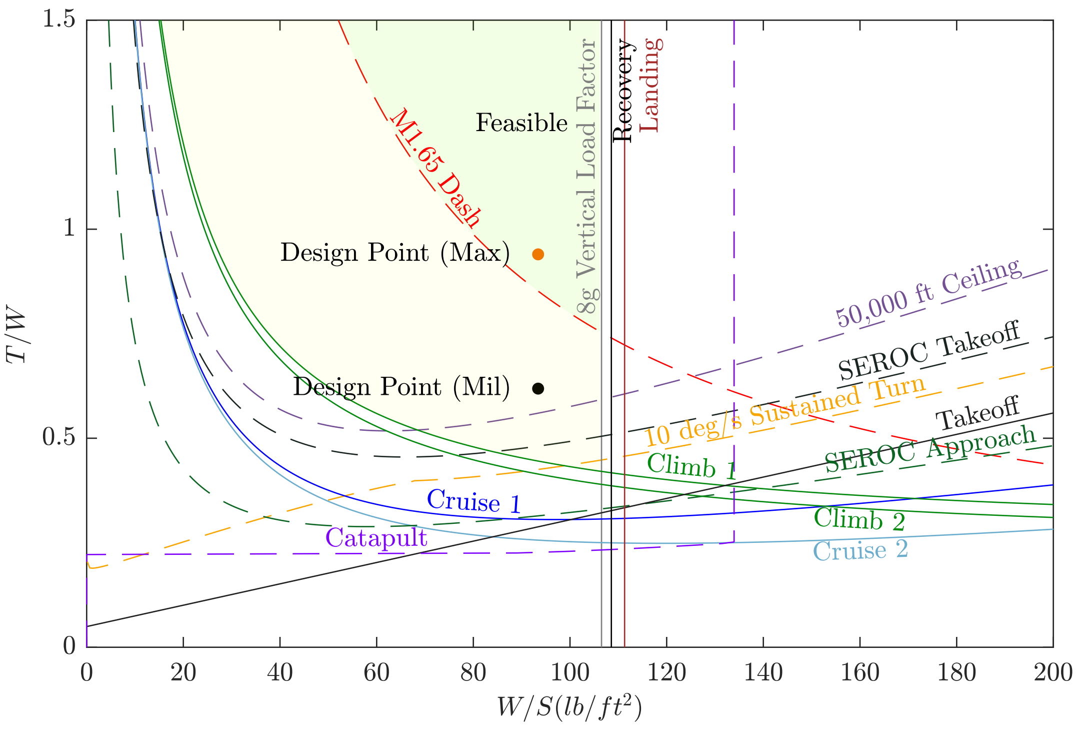

Figure 24: Validation T/W-W/S plot for the air-to-air mission based on F/A-18E performance.

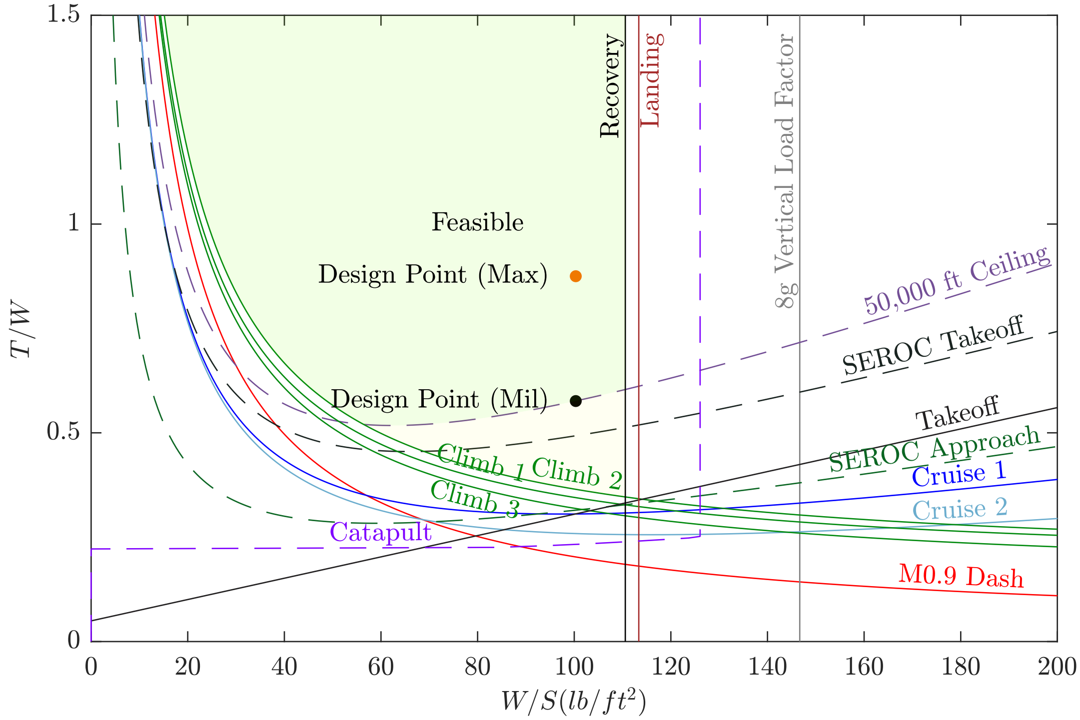

Figure 25: Validation T/W-W/S plot for the strike mission based on F/A-18E performance.

13 Objective

The objective of the RFP is to maintain comparable unit cost while delivering a fighter with higher performance than the legacy aircraft. As seen in Table 1, the F/A-67 meets these requirements, having over double the combat radius while maintaining a lower flyaway cost for 500 units.

There is no explicit objective function used, but details on design point selection for mission performance is described in §14. Details on how the aircraft unit cost is calculated are found in §17. Cost is treated as a constraint, and can be loosely considered to be part of an objective function. The flyway cost for the F/A-67 is 98.8 \$M 2025 which is less than the 113 \$M 2025 unit cost for the F/A-18E/F.

14 Design Refinement

Post-PDR design refinement involved improving various models, and the usage of several trade studies to modify the aircraft design point. The improvements to analysis models include a thorough component weight estimation for the aircraft empty weight (§6), and the implementation of a component drag estimation model for parasitic drag (§8). Another key improvement was in the fuel fraction estimation. The mission profiles were modified to utilize optimal cruises and climbs to reduce fuel burn and reduce takeoff weight.

Two trade studies performed after PDR included trading the storage of ordnance internally and externally, as well as the selection of various mission performance parameters to meet the objective of maximizing performance, while maintaining the cost requirement.

14.1 Ordnance and Fuel Tanks Trade Study

External ordnance and fuel tank drag are estimated using drag buildup methods given in Raymer [5]. For external store analysis, the aircraft is assumed to have two pylons on each wing with pylons on the wingtip to carry an AIM-9X. Each wing pylon would carry either 2 MK-83 JDAMs or 2 AIM-120Cs, with the last 2 AIM-120Cs on the fuselage. The results of the trade are given in Table 16 for the subsonic cruise condition.

Table 16: Change in drag counts for external ordnance and fuel tanks, accounting for pylons and racks.

	0 External Fuel Tanks	2 External Fuel Tanks
0 External Weapons	0	17.39
2 AIM-9X	1.31	18.70
6 AIM-120C	21.94	39.33
4 MK-83 JDAM	22.99	40.28
Air-to-Air	23.25	40.64
Strike	24.19	41.59
Fuselage (internal)	74.50	-

The results show that any additional externals significantly increase the drag. A full air-to-air loadout with 2 external fuel tanks adds over half the drag of the fuselage designed for internal storage. To counteract the additional drag from externals, the fuselage drag contribution would have to be significantly reduced, which is not feasible for this aircraft. For instance, fuselage surface area reduction would be challenging due to the fixed volume of the twin-engine configuration. Additionally, any externals will heavily penalize wave drag in the supersonic regime, as seen in Raymer [5].

Internal storage will result in a lower drag count for the whole aircraft and decrease RCS, as desired by the RFP. As such, the F/A-67 has fully internal storage for the air-to-air and strike mission.

14.2 Mission Performance and Sizing Trade

Aircraft wingspan, planform area, and AR were reanalyzed using refined weight estimation methods for empty weight prediction and fuel fraction. Additionally, detailed drag polars based on the geometry of the preliminary F/A-67 design were incorporated into the weight estimation and sizing plot routines. Because of the refined methods, the feasibility of the designs for various mission performance parameters, like combat radius, changed. The trade study was performed by varying combat radius, combat time, and dash Mach number for the air-to-air mission. The air-to-air mission was by far the more constraining mission type because of the supersonic dash and sustained-turn rate constraints. Essentially, a mission that was feasible for the air-to-air mission was feasible for the strike mission, but not the other way around. The objective analyzed for the trade study was aircraft unit cost. The specific requirement was for the unit flyaway cost to not exceed that of the F/A-18E/F (113 $M 2025). This enables a higher performing aircraft despite the increasing cost.

In addition to varying the air-to-air mission parameters, the aircraft wingspan and AR were varied over ranges based on historical aircraft. Wingspan was varied from 45 ft to 60 ft, the maximum value, and AR was between 2.7 and 4. For selecting a design point, an additional goal was to reduce wingspan and planform area over the preliminary design of 60 ft and 670 ft^2. The result of the trade study was visualized using a matrix of lattice plots, with combat radius and combat time varied in individual carpet plots, Mach number varied over a lattice plot, and wingspan and AR varied over the matrix of plots. For clarity, only relevant lattice and carpet plots are included.

The wingspan and AR were determined by identifying the lattice plot with the largest feasible region while reducing wing area from the preliminary design. The purpose of this was to reduce planform configuration issues, primarily the 50 ft length limit of the aircraft. The final wingspan and AR selected were 48.75 ft and 3.67, respectively. Using the lattice plot, a dash Mach number of 1.65 was achieved. Figure 26 shows the carpet plot for the selected wingspan, AR, and dash Mach number. The blue hatched curve separates infeasible and feasible designs using the sizing plot and constraint curves. Feasible designs are on the opposite side of the hatch marks. For all designs analyzed, the unit flyaway cost does not exceed the 113 $M 2025 requirement. An interesting trend is that for moderate combat radii and combat times, the cost is nearly constant. Eventually, increasing combat time and combat radius yield a steady increase in cost.

The selected design point was determined based on achieving at least one desired mission performance parameter. Additionally, all design points were analyzed for the desired sustained turn rate of 10 deg/s. In this case, the desired combat radius of 1,000 nmi can reasonably be reached without significantly sacrificing combat time. The desired combat time of 5 min is beyond the analyzed region of the carpet plot, but it can be extrapolated that nearly all of the combat radius would need to be sacrificed. Furthermore, meeting the desired combat radius has a lower weight penalty than combat time, resulting in a lower unit cost. Therefore, the selected mission performance is a 1,000 nmi combat radius, a 3 min combat time, a dash Mach number of 1.65, and a sustained turn rate of 10 deg/s. All parameters exceed the minimum requirements, and the combat radius and sustained turn rate meet the desired values. Combat time is increased up to the feasible limit as seen in Figure 26.

The results of the sizing trade, namely the wingspan and AR, were subsequently used for the planform trade, which is discussed in §8. After redesigning the aircraft planform and fuselage, the trade study was run again to further refine the design. This resulted in an identical wingspan and AR, but with a slightly higher air-to-air combat time of 3.5 min.

14.3 Design Changes and Finalized Sizing Plots

Since PDR, the F/A-67 has significantly changed, most visibly in the aircraft shape. The refined sizing process and subsequent trade studies reduced the aircraft wingspan, planform area, and AR compared to the preliminary design. However, to better accommodate internal fuel and payload, the fuselage width was increased. Furthermore, landing gear was included for CDR and the refined empty weight estimation code resulted in a lower empty weight. Finalized T-S plots for the F/A-

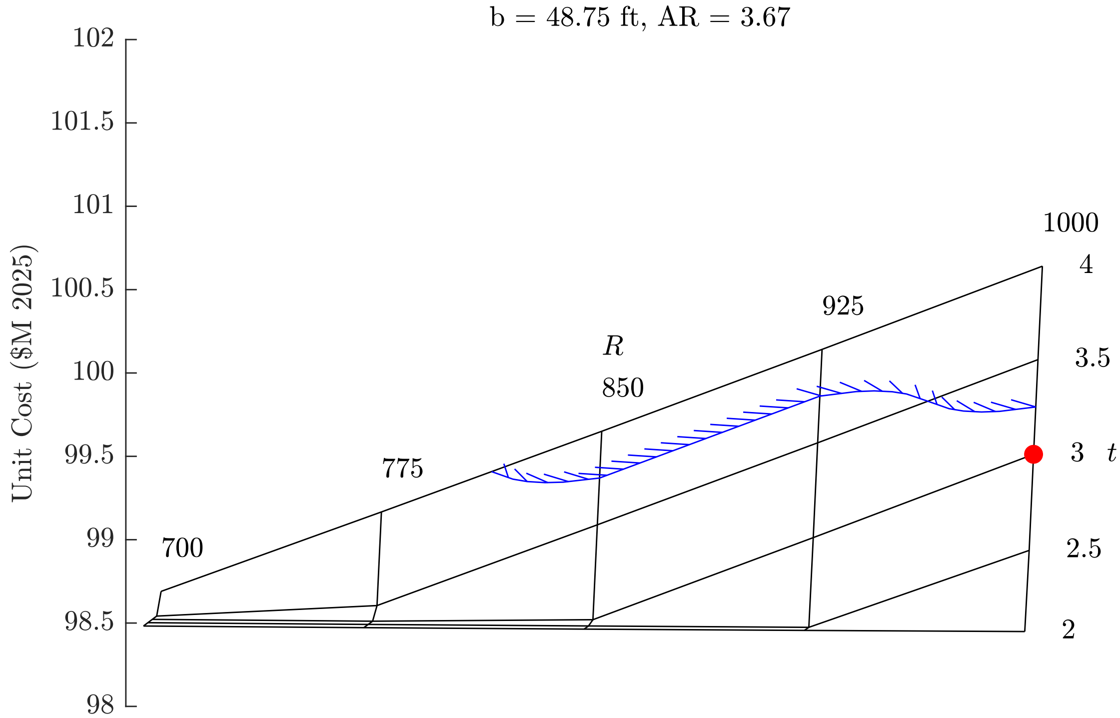

Figure 26: The carpet plot for a dash Mach number of 1.65 varies combat radius in nmi and combat time in min. The blue hatched curve roughly indicates the boundary between feasible and infeasible designs. The selected design point is marked in red.

67 are included for the air-to-air (Fig. 27) and strike (Fig. 28) missions. The design points for maximum thrust are shown to be in the feasible region. Military thrust is also indicated as is feasible. An important change between the PDR and CDR sizing is that the conventional takeoff constraint is now considered to be a military thrust constraint (solid line).

15 Structures

15.1 Load Paths

The internal structure takes the loads that aircraft will experience. The aircraft's internal structure consists of a 3-spar configuration in the wingbox, stabilators, and vertical tails. The fuselage consists of 8 ring-frames spaced to fit around the twin-engines as shown in Figure 29 and connected to wing spars. Additionally, longerons are paired normal to the frames along the length of the aircraft. The internal structure of the aircraft withstands torsional, buckling, and bending loads. The frames and longerons are also designed to fit around the ordinance loadouts and landing gears.

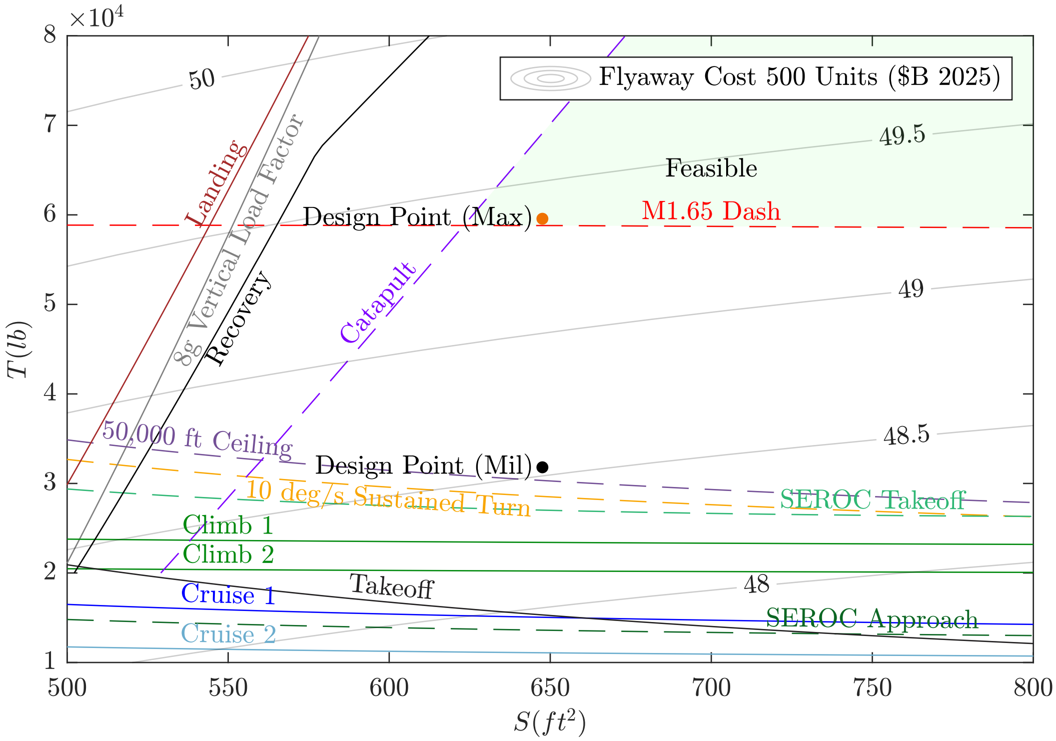

Figure 27: Final T-S plot for the air-to-air mission with contours of flyaway cost for 500 units at a given maximum thrust.

15.2 V-n Diagrams

The V-n diagrams for maximum weight and mid-mission weight (Fig. 30) indicate all relevant speeds (stall, rough air, corner, cruise, dive). Gust loads are plotted as dash-dotted lines. The ultimate maneuver load factor is plotted as a dotted line. The solid black lines indicate the combined flight envelope of limit maneuver loads and gust loads. The aircraft meets the desired design vertical load factor of 8g. Aircraft empty weight estimation also takes into account a 1.5 safety factor on the design load factor, which indicates the aircraft sizing accounts for the structural weight required to withstand the combined loads envelope.

Since the combined load envelope does not exceed the ultimate load factor (including gusts), and the aircraft weight is sized using the ultimate load factor, the team is confident that the F/A-67 structure can withstand the design maneuver loads for all flight conditions. Details regarding the calculations of the maneuver and gust envelopes is discussed Appendix F. Appendix F inclues Figure 35, a V-n diagram at minimum weight.

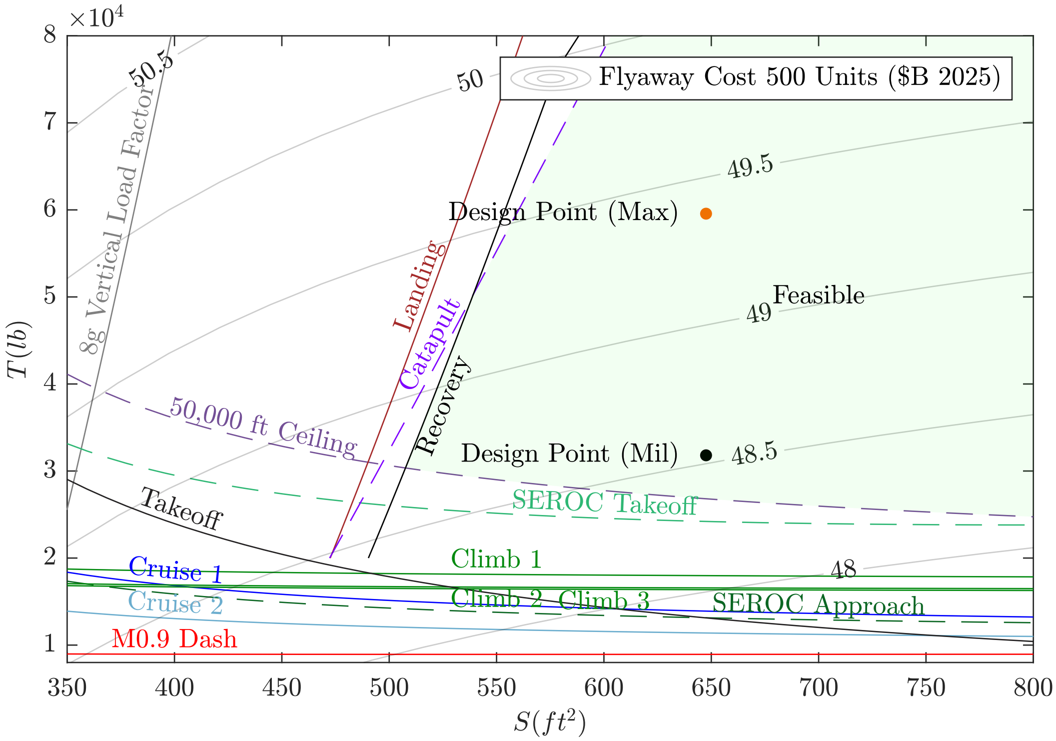

Figure 28: Final T-S plot for the strike mission with contours of flyaway cost for 500 units at a given maximum thrust.

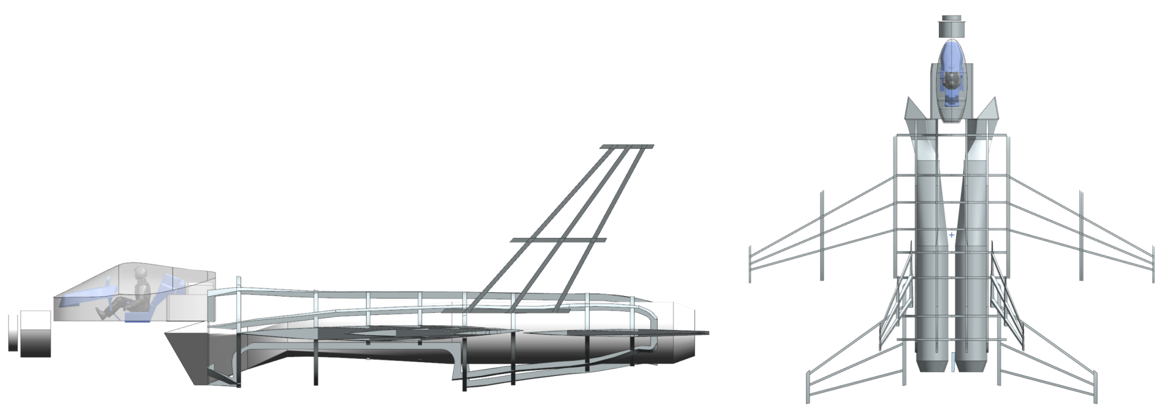

Figure 29: Aircraft layout showing load bearing internal structures.

16 Combat Performance and Maneuverability

Air-to-air combat performance is evaluated using energy maneuverability methods described in [5]. The RFP specifies maneuverability requirements for sustained turn rate at 20,000 ft, and that during air-to-air combat, the aircraft must fly at the speed for best turn. This is assumed to be either the corner speed (maximum turn rate), or the speed for best sustained turn rate. Figure 31

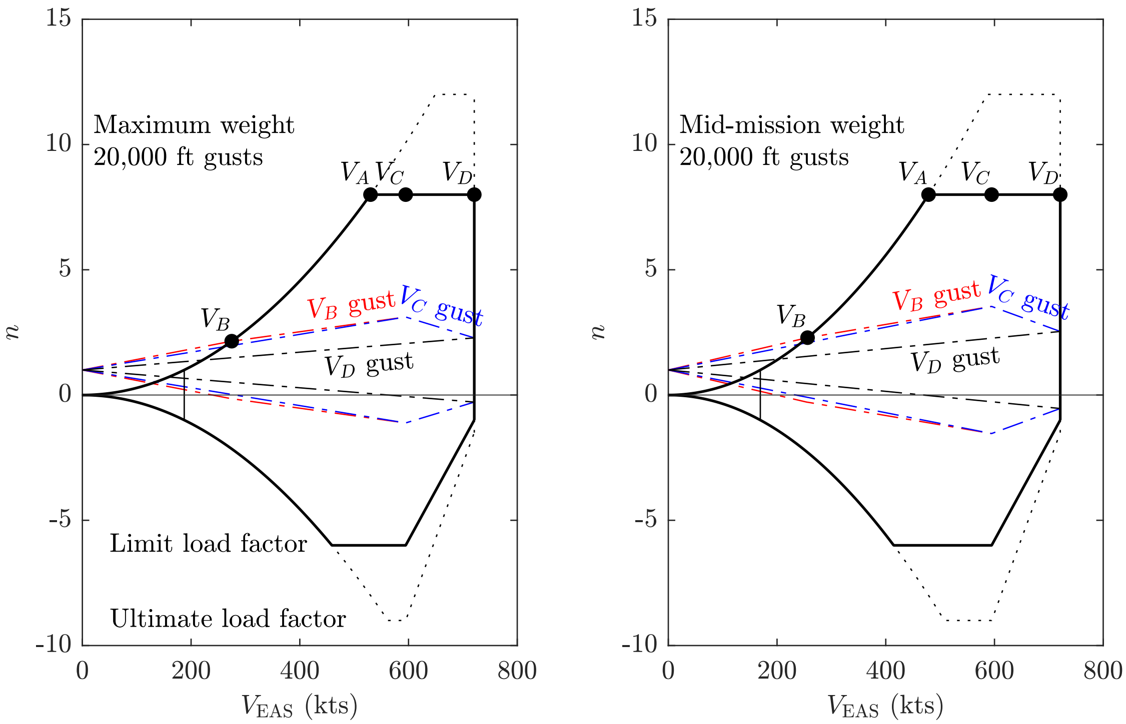

Figure 30: The V-n diagram at maximum weight and mid-mission weight. The solid black lines are the combined loads, the dotted line is the ultimate maneuver load, and the dash-dotted lines are gust loads. The 1g stall speed is denoted at the thin vertical line.

shows that the F/A-67 has a maximum sustained turn rate of roughly 12 deg/s, which exceeds the RFP requirement. This is because the sustained turn rate falls out during the design process, and the desired value of 10 deg/s is enforced by the sizing plot constraints. The maneuver load factor is 7g, which is less than the 8g limit load factor, so it is also structurally feasible.

During combat at 10,000 ft, energy maneuverability analysis (Fig 32) indicates that the aircraft has a corner speed of 536 kts at Mach 0.84 and a maximum instantaneous turn rate of 15 deg/s. The maximum sustained turn rate is 13 deg/s at Mach 1.03 or 657 kts. The load factor at these conditions is the limit load factor of 8g, making it again structurally feasible. The RFP does not explicitly specify combat turn rates or turn speeds, so the aforementioned performance parameters also fall out during the analysis process.

16.1 Air Load Distributions

The spanwise distribution of lift was calculated using AVL [8]. The $C_{L,\max}$ was used, as this is the worst case for loading. The clean loading plot (Fig 33) is the worst case for loading, as the 8g limit load factor occurs at this configuration. Additional lift distributions are shown in Appendix I.

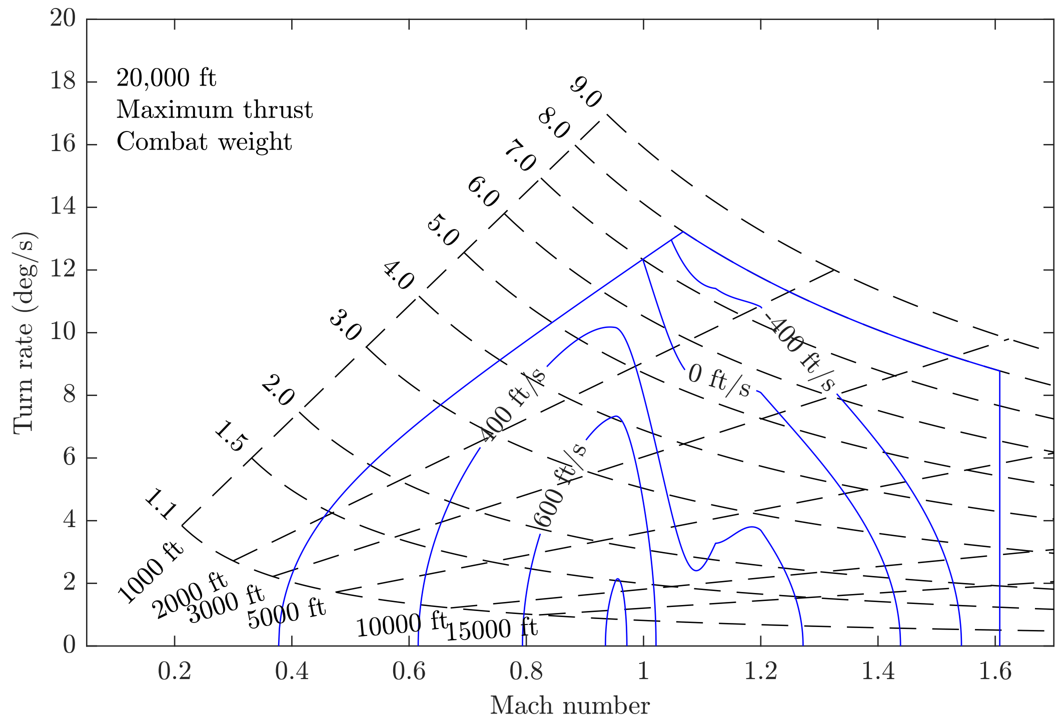

Figure 31: Energy maneuverability diagram at combat weight for the air-to-air mission at 20,000 ft. The F/A-67 exceeds the 10 deg/s desired sustained turn rate.

17 Cost

To calculate the total flyaway cost, equations from Roskam [30] were used to calculate manufacturing cost and profit. The manufacturing cost includes the total airframe engineering and design cost, total program production cost, the production flight test operations cost, and financing. The total program production cost comes out to be 36 $B 2025 as shown in Table 17. The unit engine cost is 18.8 $M 2025.

Table 17: Total program production cost.

	Cost ($B 2025)
Engine Cost	9.4
Avionics Cost	14
Manufacturing Labor Cost	7.5
Manufacturing Material Cost	2.8
Tooling Cost	1.8
Manufacturing Quality Control Cost	0.97
Total	36

The unit flyaway cost is calculated by adding the airframe engineering and design cost, program production cost, production flight test operations cost, financing cost, and profit, shown in Table 18.

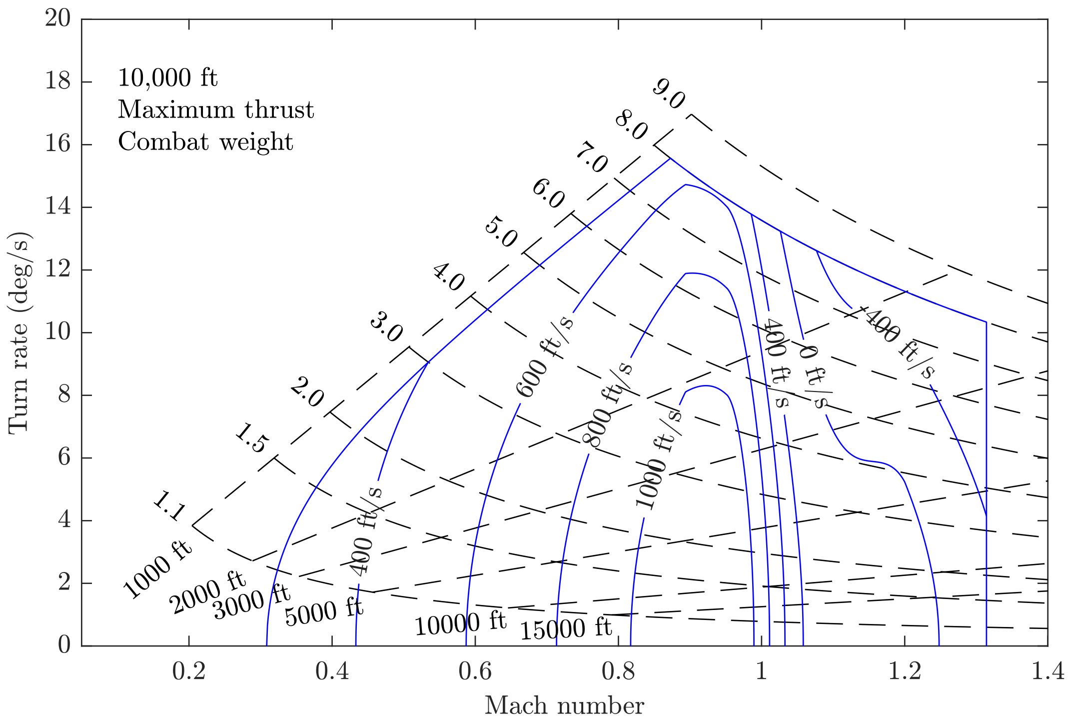

Figure 32: Energy maneuverability diagram at combat weight for the air-to-air mission at 10,000 ft.

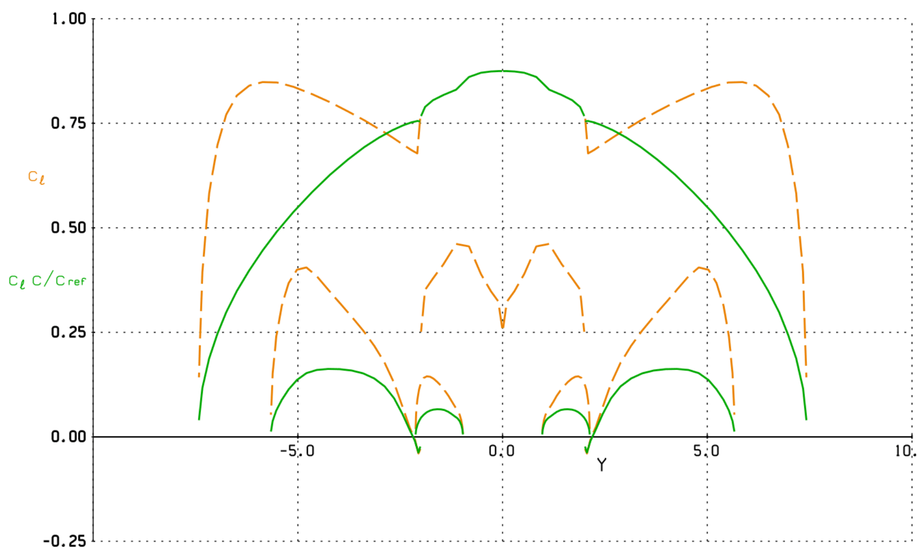

Figure 33: Clean spanwise lift distribution. The x-axis is wing y location in meters, and the y-axis is C_ℓ (orange) and $C_\ell \times c/c_{ref}$ (green).

To find unit cost, the total flyaway cost is then divided by the total number of units. The cost calculation code is discussed further in Appendix D.

Table 18: Unit flyaway cost.

	Cost ($M 2025)
Airframe Engineering and Design Cost	72.8
Program Production Cost	1.89
Production Flight Test Operations	1.66
Financing	13.5
Profit	8.98
Unit Flyaway Cost	98.8

17.1 Operating Cost

Operating costs includes the program cost of fuel, oil, and lubricants; program cost of direct personnel; cost of consumable materials; cost of indirect personnel; cost of spares; depot cost; and miscellaneous cost. The payload comes from the ordnance discussed in §10 with an assumed 50-50 mission mix. The total direct operating cost per payload nautical mile comes out to be 0.014 $/lbm-nmi for 500 units as shown in Table 19.

Table 19: Direct operating cost per payload nautical mile.

	Cost ($/lbm-nmi 2025)
Fuel	0.0021
Oil	0.000011
Aircraft Maintenance	0.0042
Engine Maintenance	0.0015
Crew	0.0023
Surface Treatments	0.00042
Miscellaneous Cost	0.0033
Total	0.014

Additional costs that are not mentioned in §17 but are important to consider are discussed in Appendix D.

17.2 Objective Function

For the objective function overlaid onto the plots in §4, the cost equations from Roskam are used [30]. The objective itself is further discussed in §13.

18 Computation Procedure

The computation procedure for configuring the F/A-67 design is an iterative process as outlined in the eXtended Design Structure Matrix (XDSM) (Fig. 34). Specifics of the various phases of the iterative process have been refined since PDR.

1. Aircraft design parameters and mission performance parameters are used as inputs for the takeoff weight estimation. Aircraft weights and mission segment weight fractions are the outputs.
2. T/W-W/S plots are used to verify that a design is feasible. Based on the constraints, the design is revised to either increase performance or bring the design into the feasible region.
3. Wing planform sizing is performed by minimizing drag at the cruise operating condition. This requires sectional airfoil drag polars for parasitic drag calculation. Airfoil design involves shape optimization for the cruise and air-to-air dash conditions. The results are fed into the planform sizing procedure which produces new operating conditions for the airfoil. Airfoil optimization and planform sizing are iterated to get a more optimal design. Wing planform analysis is performed in Athena Vortex Lattice (AVL) and airfoil optimization is done using the MACH-Aero framework.
4. Tail sizing is performed after the wing planform sizing. Analysis on static stability, including the aircraft neutral point, is found using AVL and AeroSandbox AeroBuildup.
5. Weight estimation results are used to calculate CG. The results of this are used for landing gear sizing. Other elements of the CAD feed back into weight estimation and sizing
6. The full configuration is compared against all RFP constraints, with aircraft dimension limits being the primary influence on wing planform and tail sizing. Designs are manually iterated to maximize mission performance while remaining within all sizing, design, and cost constraints.

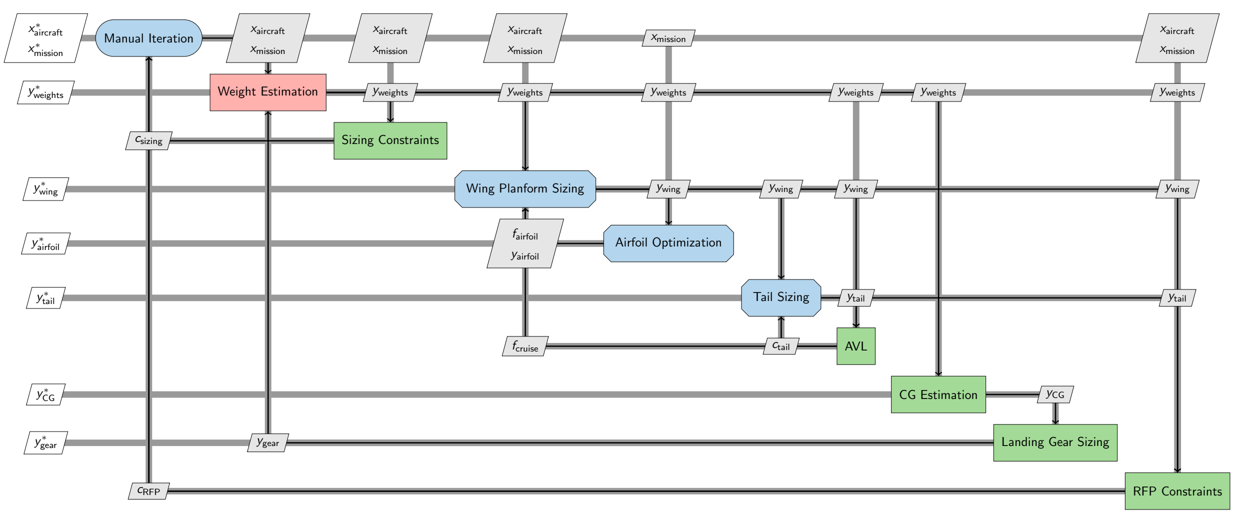

Figure 34: XDSM for the execution of the F/A-67 iterative design process.

19 Software Design

Scripts used for the F/A-67 design were created primarily in MATLAB. Source control was utilized during the process to enable parallel development of the preliminary sizing code, cost prediction code, and aerodynamics trades.

19.1 Sizing Code

The sizing code is centered around manipulating a single aircraft struct which contains information related to general aircraft parameters, like geometry or cruise performance, drag polars, and engine performance. Furthermore, the struct contains data on mission specific parameters like combat radius as well as aircraft weights. The rest of the code is separated by purpose, including weight estimation, mission profiles, mission segments, constraint functions, T-S plot solvers, drag polar estimation, and various helper functions and regressions. These functions are called in high level driver codes which handle the generation and plotting of T/W-W/S plots and T-S plots.

The algorithm used for the weight estimation code is a modification to Algorithm 1 from Martins. Equation (12) is the residual equation used to solve for the takeoff weight, W_0. The empty weight fraction, W_e/W_0, and the fuel fraction, W_f/W_0, are determined using regressions and a mission segment buildup respectively. MATLAB's `fsolve` is used to determine W_0.

$$R = W_0 - (W_{\text{crew}} + W_{\text{payload}}) - W_0 \left(\frac{W_e}{W_0} + \frac{W_f}{W_0} \right) \tag{12}$$

The weight estimation code takes for input a function handle for either the air-to-air or the strike mission. This enables the code to switch between mission types. Other algorithms derived from Martins also use `fsolve` to improve convergence and decrease runtime.

Refined empty weight estimation procedures replace the calculation of empty weight fraction for CDR. Furthermore, the refined fuel fraction estimation methods increase the code runtime. MATLAB's parallel computing toolbox is utilized extensively to increase efficiency. This enabled the development of large trade studies to refine the F/A-67 design.

19.2 Cost Prediction Code

The cost prediction code uses equations from Roskam to compute the cost, which accounts for learning rates associated with the manufacturing process. The script takes in takeoff weight, fuel weight, and block time from the initial sizing scripts to compute the costs using labor rates from Raymer. The script is organized into four main sections: auxiliary; environmental, fleet, compatibility, and weapons; direct operating cost; and aircraft unit cost. The first three sections do not contribute to the flyaway cost. However, the environmental, fleet, compatibility, and weapons section includes avionics cost, which is factored into the flyaway cost.

The aircraft unit cost is further broken down into seven subsections: labor cost, materials cost, tooling cost, quality control cost, avionics and engine cost, RDT&E cost, and totals. These subsections have an associated RDT&E cost, which is subtracted out when computing flyaway cost, as the military does not include RDT&E in flyaway costs. The subsection costs are then added together, along with financing and profit considerations, to give the final total cost and unit flyaway cost. Inflation is calculated using an inflation lookup table from 1970 to 2025 using data from the U.S. Bureau of Labor Statistics.

20 Conclusions

No Thai Grubbin's comprehensive iterative sizing and design process has produced a capable and affordable carrier-based strike fighter. The F/A-67 is sized with two powerful engines, with increased combat performance for both the air-to-air and strike missions. Despite the aircraft's large size and spot factor, the F/A-67 boasts a high combat radius of 1,000 nmi, with mission performance exceeding minimum requirements, while maintaining a lower unit cost than the F/A-18E/F. At only 98.8 $M 2025, the design satisfies all NGCSF requirements. The No Thai Grubbin team is confident in their design, factoring in feedback from PDR for improved weight modeling, aerodynamic analysis, and structural design. The F/A-67 will deliver a competitive capability required by the United States Navy for continued naval aviation dominance.

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A Glossary

Acronyms

AR aspect ratio. 13, 14, 32, 33, 42, 45, 46

L/D lift-to-drag ratio. 13, 15

T/W thrust-to-weight. 7, 14–19, 40, 41, 43, 54, 55

W/S wing loading. 7, 16–18, 43, 54, 55

ACLS automatic carrier landing system. 37

ADVEW Advanced Electronic Warfare. 37

AESA actively electronically scanned array. 37

AIAA American Institute of Aeronautics and Astronautics. 7

AVL Athena Vortex Lattice. 54

CDR Critical Design Review. 7, 14, 46, 47, 55

CG center of gravity. 6, 26, 27, 32, 39, 54

FBW fly-by-wire. 37

IDECM Integrated Defensive Electronic Countermeasures. 37

JPALS joint precision approach and landing system. 37

LE leading edge. 30, 33–35

MAC mean aerodynamic chord. 6, 27, 28

NGCSF Next Generation Carrier-Based Strike Fighter. 7, 11, 56

OBOGS On-Board Oxygen Generation System. 38

PDR Preliminary Design Review. 7, 14, 37, 44, 45, 47, 53, 56

RCS radar cross-section. 14, 36, 38, 45

RDT&E Research, Development, Test & Evaluation. 6, 55, 69

RFP Request for Proposal. 7, 11–17, 19, 20, 25, 33, 36, 37, 40–42, 44, 45, 49, 50, 54

SEROC single-engine rate of climb. 13, 17, 40, 41

SM static margin. 26–28, 32, 37

TE trailing edge. 30, 33–35

TRL Technology Readiness Level. 11, 37

TSFC thrust specific fuel consumption. 36

WOD wind over deck. 13, 40, 41

XDSM eXtended Design Structure Matrix. 53

Nomenclature

$\Delta_{c/4}$ Quarter chord sweep angle

Δ_{LE} Leading edge sweep angle

$\dot{\psi}$ Yaw rate

ℓ_{HT} Horizontal stabilizer moment arm length

η Shock absorber efficiency

η_T Tire shock absorber efficiency

γ Ratio of specific heats

λ Taper ratio

ρ Air density

θ_{twist} Twist angle

AR Aspect ratio

c Thrust specific fuel consumption

c_ℓ Sectional lift coefficient

C_D Drag coefficient

c_d Sectional drag coefficient

c_f Skin friction coefficient

C_L Lift coefficient

C_m Pitching moment coefficient

C_n Yaw moment coefficient

c_p Pressure coefficient

$c_{\ell,\max}$ Maximum sectional lift coefficient

C_{D0} Parasitic drag coefficient

C_{Di} Induced drag coefficient

C_{Dp} Wing parasitic drag

C_{Dw} Wing wave drag

c_{HT} Horizontal tail volume coefficient

$C_{L,\max}$ Maximum lift coefficient

$C_{l\beta}$ Derivative of roll moment coefficient with side-slip angle

$C_{m\alpha}$ Derivative of pitching moment coefficient with angle of attack

$C_{n\beta}$ Derivative of yaw moment coefficient with side-slip angle

c_{VT} Vertical tail volume coefficient

D_{main} Main gear tire diameter

E Endurance

e Oswald efficiency number

g Gravitational acceleration

h_{stroke} Landing gear stroke length

K $1/(\pi ARe)$

KE_{breaking} Kinetic energy dissipated on landing

L/D Lift-to-drag ratio

$L/D_{\max}$ Maximum lift-to-drag ratio

M Mach number

M_{crit} Critical Mach number

n Load factor

N_{gear} Gear load factor

n_z Vertical load factor

q Dynamic pressure

R Range

S Wing reference area

S_{wet} Wetted area

S_T Tire stroke length

t/c thickness-to-chord ratio

T/W Thrust-to-weight ratio

V Velocity

V_γ Climb rate

V_{stall} Stall speed

V_{vertical} Sink rate

W/S Wing loading

W_0 Takeoff weight

W_{landing} Landing weight

W_{main} Main gear tire width

W_i Weight at the i^{th} mission segment

W_W Static weight on wheel

x_{CG} x-coordinate of

B Sizing Constraints

B.1 Takeoff

Given a maximum allowable ground roll, s_{tog}, thrust to weight in terms of wing loading can be solved for using Eq. (13):

$$s_{tog} = \frac{k_1(W/S)_{TO}}{\rho[C_{L_{maxTO}}(k_2(T/W)_{TO} - \mu_G) - 0.72C_{D_o}]} \tag{13}$$

where $k_1 = 0.0447$, μ_G is the rolling coefficient of friction, and k_2 is given by Eq. (14):

$$k_2 = 0.75(\frac{\lambda + 5}{\lambda + 4}) \tag{14}$$

where λ is the engine bypass ratio.

B.2 Landing

According to Roskam, the landing field length is given by Eq. (15):

$$s_{fl} = 0.3 * V_A \tag{15}$$

where s_{fl} is given in feet and V_A, the approach speed, is in knots. Approach speed is given by Eq. (16):

$$V_A = 1.2 * V_{SL} \tag{16}$$

where V_{SL} is given by Eq (17). V_{SL} must be converted to knots.

$$V_{SL} = \sqrt{\frac{2(W/S)}{\rho * C_{L_{max}}}} \tag{17}$$

B.3 Catapult Launch

The RFP sets the catapult minimum end speed using four requirements. They are repeated here.

1. An end speed which results in the CG position of the aircraft sinking no more than 10 feet from its position at the end of the power stroke, with a deck run not to exceed 32 feet (distance from the end of the power stroke to round-down), with cockpit control position held either fixed or free or controls active.
2. The speed represented by 90% of the maximum lift coefficient, power off, out of ground effect.
3. The minimum airspeed at which the aircraft has a longitudinal acceleration of 0.065 g (2.0913 ft/sec2) at zero flight path angle.
4. If multi-engine, the minimum aircraft control speed with one engine inoperative.

The first requirement is interpreted as limiting the distance the aircraft can drop after flying past the round-down. The aircraft reaches its end speed at the end of the power stroke before accelerating for the 32 ft deck run. The aircraft then flies off the carrier deck.

The second requirement end speed can be calculated as a function of W/S.

$$V_{\text{end},2} = \sqrt{\frac{2}{\rho \cdot 0.9 C_{L_{\max}}} \frac{W}{S}} \tag{18}$$

Assuming that the aircraft can quickly achieve $0.9C_{L_{\max}}$ as it leaves the deck, the aircraft will be in level flight since it's velocity will be at least $V_{\text{end},2}$. Thus, the first requirement is considered to be satisfied since the aircraft will not sink after leaving the deck. This implies that the end speed that satisfies the first requirement is less than $V_{\text{end},2}$.

For the third requirement, level flight is assumed, with longitudinal acceleration being a minimum of 0.065g.

$$\frac{T-D}{W} = 0.065 \quad \text{and} \quad L = W \tag{19}$$

Expanding the expressions in (19) and combining them together yields an expression for minimum airspeed as a function of W/S, T/W, and other aircraft parameters.

$$\frac{T}{W} - \frac{\rho V^2 \left(C_{D_0} + K \left(\frac{2}{\rho V^2} \frac{W}{S} \right)^2 \right)}{2(W/S)} = 0.065 \tag{20}$$

The minimum airspeed can be found by solving equation (20) for a specified W/S and T/W.

The minimum aircraft control speed for the fourth requirement is interpreted to be the minimum speed where a SEROC of 200 ft/s can be achieved at takeoff. Since the maximum takeoff weight allowed by the RFP is 90,000 lb, the minimum speed is set to 131 kts. This is found using the catapult performance curves provided in the RFP. If the SEROC requirement is satisfied, then the minimum speed for aircraft control will be less than or equal to the velocities specified from the second and third requirements for catapult end speed.

B.4 Recovery

The RFP defines the arrestment engaging speed to be 5% greater than the approach speed. The approach speed is required to 10% above stall speed, but shall be less than 145 kts. An additional point requirement lists the maxiumum wind-over-deck (WOD) to be 15 kts. A relationship between stall speed and arrestment speed can be defined as

$$V_{\text{stall}} = V_{\text{eng}}/1.05/1.1 + 15 \text{ kts} \tag{21}$$

Eq. (21) assumes that all speeds (engaging, approach, and stall) are defined as relative speeds. Thus, the true aircraft stall speed is 15 kts faster due to the minimum WOD. The equation can be substituted into the constraint equation for aircraft stall.

$$\frac{W}{S} = \frac{1}{2}\rho \left(\frac{V_{\text{eng}}}{1.05 \times 1.1} + 15 \text{ kts} \right)^2 C_{L_{\max}} \tag{22}$$

The maximum allowable engaging speed is found using the arresting gear performance curves provided in the RFP. The engaging speed is a function of the landing weight and increases as the weight decreases. At 40,000 lb, the engaging speed is capped to a maximum value of 145 kts. This satisfies the RFP's approach speed requirement.

C Thrust Offtakes

Thrust offtakes are calculated using Raymer [5] and Nicolai [3].

C.1 Inlet Pressure Loss Thrust Correction

The percent thrust loss is given by Raymer and Nicolai and is calculated using Eq. 23:

$$\%_{TL_{PR}} = c_r[\eta_{\text{ref}} - \eta_{\text{actual}}] * 100\% \tag{23}$$

where c_r is the inlet ram recovery correction factor and given by Eq. 24 if manufacturer's data is not available:

$$\begin{cases} c_r = 1.35 & M_\infty \leq 1 \\ c_r = 1.35 - 0.15(M_\infty - 1) & M_\infty > 1 \end{cases} \tag{24}$$

The term eta_{actual}, actual internal pressure recovery, is given by Raymer as 0.94 for an S-duct inlet.

The term eta_{ref} is dependent on the Mach number. To find an equation relating the Mach number to the actual internal pressure recovery, a regression must be performed based on numbers given by MIL-E-5008 [31]. Raymer and Nicolai give an equation based on this MIL specification, but the implementation of that equation gives an imaginary value for η_{actual} and as such, is not used. The points given by the MIL-E-5008 are given in Eq. 25:

$$\begin{cases} 0.95 & 0 < M \leq 1 \\ 0.85 & M = 2 \\ 0.60 & M = 3 \end{cases} \tag{25}$$

By fitting a quadratic in MATLAB to the points above, eta_{actual} is given by Eq. 26

$$\begin{cases} eta_{\text{ref}} = 1 & 0 < M_\infty \leq 1 \\ eta_{\text{ref}} = -0.0750M_\infty^2 + 0.1250M_\infty + 0.9 & M_\infty > 1 \end{cases} \tag{26}$$

C.2 Engine Bleed

The percent thrust loss is calculated using Eq. 27:

$$\%_{TL_{EB}} = C_b \frac{\dot{m}_b}{\dot{m}_e} \tag{27}$$

where C_b is the bleed correction factor and is assumed to be 2 as given by Raymer and Nicolai. The bleed mass flow to engine mass flow ratio is typically between 1 and 5%. Roskam part VI [32] gives a value of 0.04 for attack jet fighters so that is the bleed mass flow to engine mass flow ratio used for the thrust correction factor.

C.3 Installed Thrust

Installed thrust is calculated using Eq. 28.

$$T_{\text{installed}} = T(1 - \frac{\%_{TL_{PR}} + \%_{TL_{EB}}}{100}) \tag{28}$$

Both thrust corrections are assumed to be reduced separately from the given thrust, rather than by multiplying one correction by the other correction by the thrust. This results in a thrust loss of approximately 10%, which is accounted for in the sizing code.

D Cost Equations

. All cost equations are taken from Roskam, with labor rates from Raymer.

D.1 Research and Development Costs

$$RDTE_{unit} = (C_{ead_r} + C_{dst_r} + C_{fta_r} + C_{fto_r}) * (1 + F_{pro_r} + F_{fin_r}) \tag{29}$$

where C_{ead_r} is Engineering and Design Cost, C_{dst_r} is Development Support and Testing, and C_{fta_r} is Flight Test Airplane Cost. F_{pro_r} represents a profit factor and F_{fin_r} represents a financing factor.

D.2 Engine(s)

According to the U.S Department of Defense [33], GE Aircraft Engines provided 110 F414 Engines for a total price 440 Million dollars for the F/A-18 E/F program. The transaction was executed in September of 2012, with each engine priced 4 million USD, now equivalent to 5.64 million USD in 2025. This aircraft will utilize two F414 Engines, resulting in a total engine price of 11.28 million USD per aircraft in 2025.

D.3 Crew and Life Support

Costs are estimated using one crew member.

Crew cost is estimated to be 2352 USD per flight. This cost is estimated using Roskam [30] Eq. (30):

$$C_{\text{crewpr}} = N_{\text{serv}} * N_{\text{crew}} * R_{\text{cr}} * Pay_{\text{crew}} * OHR_{\text{crew}} * N_{\text{yr}} \tag{30}$$

where N_{serv} is the average number of airplanes of type in service, N_{crew} is the number of crew,R_{cr} is the crew ratio per airplane, Pay_{crew} is the crew hourly rate (115 USD in 2012 or 161 USD in 2025 according to Raymer [5]), OHR_{crew} is the crew overhead rate factor, and N_{yr} is years in active service.

Life support cost includes helmets and ejection seats. Other aspects of life support systems were researched, but no credible sources were found regarding cost. Helmets cost 400000 USD per helmet in 2024, according to an article written on the Gen 3 HDMS helmet used in the F-35 [34]. Adjusting for inflation, a pilot helmet is 412002 USD in 2025. According to NAVAIR [35], the average cost of an ejection seat in 1999 was 195000 USD. Accounting for inflation, each ejection seat has an average cost of 379200 USD in 2025. This leads to a total cost of 791202 USD in 2025.

D.4 Oil

According to the U.S Secretary of Defense [36], Jet Engine Lubricating Oil, Grade 1010, has a cost of 14.58 USD per gallon in 2025. A density of 7.2 lb/gal at 15.6°C was derived from Radco Industries [37], an oil lubricant brand trusted by the U.S Department of Defense. The oil will cost 102.63 USD for the air-to-air mission and 72.72 USD per strike mission.

D.5 Fuel

According to the U.S Department of Defense [36, 38] in 2025, JP-5 fuel has a cost of 4.05 USD/gal and a density of 6.8 lb/gal at 15°C. For the air to air mission the aircraft has a fuel weight of 34465 lbs, meaning that a full tank of fuel will cost 20526 USD. For the strike mission, the aircraft has a fuel weight of 32458 lbs, resulting in a full tank of fuel cost of 19331 USD.

D.6 Ordnance

The total cost of ordnance in 2025, 7.41 million USD, comes from the following:

1. Air-to-air mission (6.34 million USD)
   (a) AIM-120C, Quantity 6
   (b) AIM-9X, Quantity 2
2. Strike mission (1.07 million USD)
   (a) MK-83 JDAM, Quantity 4
   (b) AIM-9X, Quantity 2

D.7 Avionics

The total cost of avionics is estimated to be 14$B 2025. This cost is calculated in the cost estimation code by the weight of avionics, requiring iteration to back out the weight of the avionics to get the final avionics cost. The weight of avionics is 2500 lbs as per the RFP.

D.8 Surface Treatments

Hill Air Force Base [39] reports a 61 million USD contract for corrosion control and paint for 233 F/A-18 aircraft in 1993. In 2025, this results in a price of 136.76 million USD, or 560512 USD per aircraft.

D.9 Airframe Maintenance and Engine Maintenance

$$C_{mpersdir} = C_{N_{serv}} * C_{N_{yr}} * C_{U_{ann_{flt}}} * C_{MHR_{flthr}} * C_{R_{mml}} \tag{31}$$

where $C_{N_{serv}}$ is the average number of airplanes of type in service, $C_{N_{yr}}$ is the years in active service, $C_{U_{ann_{flt}}}$ is the annual flight hours, $C_{MHR_{flthr}}$ is the maintenance man-hours per flight hour, and $C_{R_{mml}}$ is the military maintenance labor rate.

Engine maintenance costs and airframe maintenance costs are separated using an engine maintenance ratio determined from a report written by the Government Accountability Office [40]. This leads to an airframe maintenance cost of 3.13 million USD per mission and an engine maintenance cost of 1.11 million USD per mission in 2025. These costs represent the average cost per mission.

D.10 Cost Equations Results

Additional costs were investigated that do not pertain to flyaway cost, but are still important to consider.

D.10.1 Research, Development, Test and Evaluation Cost

RDT&E costs include the engineering and design cost, development support and test cost, flight test airplanes cost, flight test operations cost, profit, and financing. The labor rates used are from Raymer. The total cost comes out to be 7.4 $B for 500 units as shown in Table 20.

Table 20: RDT&E cost.

	Cost ($B 2025)
Engineering & Design Cost	1.20
Development Support & Testing Cost	0.222
Flight Test Airplanes Cost	3.83
Flight Test Operations Cost	0.302
Profit	0.740
Financing	1.11
Total	7.40

D.10.2 Environmental, Fleet Compatibility, & Weapons

Auxiliary cost includes JDAM cost, AIM-120C cost, AIM-9X cost, life support cost, avionics cost, and surface treatments cost as shown in Table 21.

Table 21: Environmental, fleet compatibility, and weapons cost.

	Air-to-Air Cost ($M 2025)	Strike Cost ($M 2025)
Total JDAM Cost	0	0.160
Total AIM-120C Cost	5.46	0
Total AIM-9X Cost Cost	1.17	1.17
Life Support Cost	0.788	0.788
Avionics Cost	28.0	28.0
Surface Treatments Cost	0.203	0.203

E Component Weight Buildup

A detailed weight component buildup is done using equations from Raymer. CG estimations of these components are also estimated, with smaller items falling into an "all else" category, from Raymer. These metrics are then used to update our empty weight and center of gravity predictions. The following sections detail the various components of the aircraft weight. All fudge factors are from Table 15.4 in Raymer, and the highest of each is assumed for a more conservative estimate.

E.1 Airframe Weight

This section will discuss the calculations needed for the weight of various parts of the airframe.

E.1.1 Wing Weight

The wing weight is found using Eq 32:

$$W_{\text{wing}} = K_{\text{comp}} * 0.0103 K_{dw} K_{vs} (W_{dg} N_z)^{0.5} S_w^{0.622} AR^{0.785} (t/c)_{\text{root}}^{-0.4} (1+\lambda)^{0.05} cos(\Lambda_{c/4})^{-1} S_{csw}^{0.04} \tag{32}$$

where K_{comp} is a composite fudge factor, K_{dw} is a delta wing factor, K_{vs} is a variable sweep factor, W_{dg} is the flight design gross weight, N_z is the ultimate load factor, given by 1.5 times the limit load factor, S_w is the trapezoidal area of the wing, AR is the aspect ratio, $(t/c)_{\text{root}}$ is the thickness to chord ratio at the root of the wing, λ is the taper ratio, $\Lambda_{c/4}$ is the sweep at quarter-chord, and S_{csw} is the wing control surface area.

The F/A-67 is not a delta wing and does not have variable sweep, so those constant factors are set to 1. As assumed in all equations, the flight design gross weight is assumed to be the empty weight of the aircraft plus 40% of the fuel weight. The limit load factor is 8. The rest of the parameters are determine using wing geometry.

E.1.2 Fuselage

The fuselage weight is found using Eq 33

$$W_{\text{fuselage}} = K_{\text{carrier}} K_{\text{comp}} 0.499 K_{dwf} W_{dg}^{0.35} N_z^{0.25} F_{\text{L}}^{0.5} F_{\text{D}}^{0.849} F_{\text{W}}^{0.685} \tag{33}$$

where K_{carrier} is a fudge factor applied for carrier aircraft, K_{dwf} is a constant for delta wing aircraft (assumed to be 1), and F_{L}, F_{D}, F_{W}, are the structural fuselage length, depth, and width respectively.

E.1.3 Horizontal Tail

The horizontal tail weight is found using Eq 34

$$W_{\text{HT}} = K_{\text{comp}} 3.316 (1 + F_w/B_{\text{HT}})^{-2} \frac{W_{dg} N_z}{1000}^{0.260} S_{\text{HT}}^{0.806} \tag{34}$$

where F_w is the width of the fuselage at the horizontal tail, B_{HT} is the span of the tail, and S_{HT} is the area of the horizontal tail.

E.1.4 Vertical Tail

The vertical tail weight is found using Eq 35

$$\begin{aligned} W_{\text{VT}} = {} & 2 K_{\text{comp}} 0.452 K_{rht} (1 + H_t/H_v)^{0.5} (W_{dg} N_z)^{0.488} S_{\text{VT}}^{0.718} M^{0.341} L_t^{-1.0} \\ & \times (1 + S_r/S_{\text{VT}})^{0.348} AR_{\text{VT}}^{0.223} (1 + \lambda_{\text{VT}})^{0.25} cos(\Lambda_{\text{VT}})^{-0.323} \end{aligned} \tag{35}$$

Where K_{rht} is a constant for a rolling horizontal tail (1.047 given by Raymer), H_t/H_v is the horizontal tail height above the fuselage divided by the vertical tail height resulting in a zero term. The rest of the terms follow the same conventions as the wing equation, with the subscript VT denoting vertical tail. The equation is multiplied by two to consider both vertical tails on the aircraft.

E.2 Engine Related Components

This section discusses the engine related components of the weight buildup, as given by Raymer. Engine selection accounts for engine offtakes, as discussed in previous assignments.

E.2.1 Engine Mounts

The engine mounts weight is calculated using Eq.36:

$$W_{\text{Engine Mounts}} = K_{\text{comp}} 0.013 N_{\text{en}}^{0.795} (N_{\text{en}} * T_{\text{e}})^{0.579} N_z; \tag{36}$$

where N_{en} is the number of engines and T_{e} is the thrust per engine in lb.

E.2.2 Engine Section

The weight of the engine sections are determined using Eq 37:

$$W_{\text{Engine Section}} = K_{\text{comp}} 0.01 W_{\text{en}}^{0.717} N_{\text{en}} N_z \tag{37}$$

where W_{en} is the weight of one engine.

E.2.3 Air Induction

The engine air induction weight is calculated using Eq. 38:

$$W_{\text{Air Induction}} = K_{\text{comp}} 13.29 K_{vg} L_d^{0.643} K_d^{0.182} N_{\text{en}}^{1.498} L_s/L_d{}^{-0.373} D_{\text{e}} \tag{38}$$

where K_{vg} is a constant for variable geometry, L_d is the air induction duct length, L_s is the length of the portion of the duct that is not split, K_d is a duct geometry constant, and D_{e} is the diameter of the engine.

K_{vg} is assumed to be one since the duct does not vary in geometry. L_s is assumed to be equal to L_d since the ducts never combine, there is only one duct for each engine. K_d is assumed to be 2.75 from Figure 15.3 in Raymer based on our inlet geometry.

E.2.4 Engine and Oil Cooling

Eq. 39 and Eq. 40 calculate engine and oil cooling weights respectively.

$$W_{\text{Engine Cooling}} = 4.55 D_{\text{e}} L_{sh} N_{\text{en}} \tag{39}$$

$$W_{\text{Oil Cooling}} = 37.82 * N_{\text{en}}^{1.023} \tag{40}$$

were L_{sh} is the length of the engine cooling shroud.

E.2.5 Engine Controls and Starter(Pneumatic)

The final components of the engines, beyond the engines themselves, are given by Eq. **??** and 42:

$$W_{\text{Engine Controls}} = 10.5 N_{\text{en}}^{1.008} L_{\text{ec}}^{0.222} \tag{41}$$

$$W_{\text{Starter(pneumatic)}} = 0.025 T_{\text{e}}^{0.760} N_{\text{en}}^{0.72} \tag{42}$$

where L_{ec} is the routing distance from engine front to cockpit, total distance if multi-engine.

E.3 Equipment

This section discusses the weight of equipment in the aircraft.

E.3.1 Flight Controls

The weight of the flight controls is calculated using Eq 43:

$$W_{\text{Flight Controls}} = 36.28 M^{0.003} S_{\text{cs}}^{0.489} N_s^{0.484} N_{\text{c}}^{0.127} \tag{43}$$

where S_{cs} is the total control surface area, N_s is the number of flight control systems, N_{c} is the number of crew.

E.3.2 Instruments

The weight of the instruments is calculated using Eq 44:

$$W_{\text{Instruments}} = 8.0 + 36.37 N_{\text{en}}^{0.676} N_t^{0.237} + 26.4(1 + N_{\text{ci}})^{1.356} \tag{44}$$

where N_t is the number of fuel tanks, currently 5 per the CAD, and N_{ci} is the crew equivalence number which is one for a single crew member.

E.3.3 Hydraulics

The hydraulics weight is calculated using Eq 45:

$$W_{\text{Hydraulics}} = 37.23 * K_{vsh} * N_u^(0.664); \tag{45}$$

where K_{vsh} is a variable wing sweep constant assumed to be one and N_u is the number of hyrdualic utility functions. Raymer estimates N_u to be between 5 and 15, given the F/A-67, and N_u of 15 was chosen.

E.3.4 Electrical

The weight of the electrical system is calculated using Eq. 46:

$$W_{\text{Electrical}} = 172.2 K_{mc} R_{kva}^{0.152} * N_{\text{c}}^{0.10} L_a^{0.10} N_{\text{gen}}^{0.091} \tag{46}$$

where K_{mc} is an electrical failure constant, R_{kva} is the system electrical rating, L_a is the electrical routing distance from generators to avionics to the cockpit, and N_{gen} is the number of generators.

The electrical failure constant is assumed to be 1.45, as given by Raymer. Also given by Raymer, the system electrical rating for fighter jets between 110-160 KvA, so it is assumed to be 160 KvA since a higher system electrical rating increases the weight of the system for a more conservative estimate. The number of generators is typically assumed to be the same as the number of engines.

E.3.5 Furnishings

The weight of the furnishings, including crew seats, is only dependent on the number of crew and is given by Eq. 47:

$$W_{\text{Furnishings}} = 217.6 N_{\text{c}} \tag{47}$$

E.3.6 Air Conditioning

The air conditioning system is dependent on the uninstalled weight of avionics. The requirement of 2500 lb from the RFP is taken to be the installed avionics weight. The uninstalled avionics weight is calculated using Eq. 48 and the air conditioning weight is calculated using Eq. 49.

$$W_{\text{Uninstalled Avionics}} = \frac{W_{\text{Avionics}}}{2.117}^{1/0.933} \tag{48}$$

$$W_{\text{Air Conditioning}} = 201.6 \frac{W_{\text{Uninstalled Avionics}} + 200 N_{\text{c}}}{1000}^{0.735} \tag{49}$$

E.3.7 Handling Gear

The handling gear is given by Eq. 50:

$$W_{\text{Handling Gear}} = 3.2 * 10^{-4} W_{dg} \tag{50}$$

E.4 Fuel System

The weight of the entire fuel system is calculated using Eq. 51:

$$W_{\text{Fuel System}} = 7.45 V_t^{0.47} (1 + V_i/V_t)^{-0.095} (1 + V_p/V_t) N_t^{0.066} N_{\text{en}}^{0.052} (T \times TSFC/1000)^{0.249} \tag{51}$$

where V_t is the total volume of the tanks, V_i is the volume of integral tanks, V_p is the volume of self-sealing tanks, T is the maximum thrust of the combined engines, and $TSFC$ is the thrust specific fuel consumption as maximum thrust. All tanks are assumed to be self-sealing for better fuel tank protection.

E.5 Avionics Weights

The avionics weight is fixed at 2500 lb. The primary avionics systems weights are detailed in Table 22. The JPALS system uses an onboard GPS to interface with the JPALS network. The H-764 is the GPS chosen as it has been proven on most military aircraft [41]. The F/A-18 Advanced Mission Computer from General Dynamics does not have a publicly available weight, but the JARVIS Mission Computer provides a good estimate [42]. There are four flight control computers as part of the quadruple redundant fly-by-wire system [43]. The rest of the avionics will be composed of other essential components. However, due to lack of information, the weights of the other components part of the 2500 lb avionics weight were unable to be found.

Table 22: Available Avionics Weights

	Weight of Avionics (lb)
AN/APG-81 Radar [44]	485.0
H-764 JPALS GPS	19.9
Network Centric Flight Control Computers	140.0
AN/ALQ-214 IDECM [45]	162.2
JARVIS Mission Computer	42.1
Total	849.2

F V-n Diagram Calculation

V-n diagrams were created for the F/A-67 at three flight conditions:

- Maximum mission weight (air-to-air)
- Mid-mission weight (air-to-air)

- Minimum mission weight (strike)

The mission profile with the heaviest takeoff weight is air-to-air, and the mission profile with the lower minimum weight is strike. Mid-mission weight was determined for the air-to-air mission due to the increased maneuverability requirements for air-to-air combat. Mid-mission weight is defined when the aircraft enters combat. The maximum positive limit load factor N_z was set using RFP requirements, and feasibility from sizing plot analyses. Aerodynamically, the aircraft can meet the desired 8g maximum positive load factor. For maximum negative limit load factor, a value of -6g was selected based on recommendations from Raymer [5].

The calculation of maneuver loads and gust loads were determined using equations from Raymer [5]. The load factor n for maximum lift was calculated as

$$n = \frac{1}{2W}\rho_{\text{sl}}V_{\text{EAS}}^2 S C_{L,\max} \tag{52}$$

where sea level density and equivalent air speed are used. The corner speed V_A was calculated using equation (53). The positive and negative limit load factors use an identical $C_{L,\max}$ value. This is assumed since the F/A-67 airfoil is nearly symmetric.

$$V_A = \sqrt{\frac{2WN_z}{\rho_{\text{sl}} S C_{L,\max}}} \tag{53}$$

The maximum cruise speed V_C was selected as the design air-to-air supersonic dash speed of Mach 1.65. The maximum dive speed V_D was selected as the desired RFP dash speed of Mach 2.0. Velocities were determined at the dash altitude of 30,000 ft and were converted to equivalent airspeeds using equation (54).

$$V_{\text{EAS}} = \sqrt{\frac{\rho}{\rho_{\text{sl}}}}V \tag{54}$$

For the V-n diagram, the negative maneuver limit load factor between V_C and V_D increases from the maximum negative load factor to a value of -1g. This was determined based on guidelines for fighter aircraft in MIL-A-8861B [46]. A safety factor of 1.5 is applied to the maximum limit load factor to give the ultimate load factor.

F.1 Gust Loads

Gust loads were calculated for rough air, at V_C, and at V_D, at 20,000 ft. The gust load factor is calculated using equations (55)-(58) [5]. The load factor is defined as

$$n = 1 \pm \frac{\rho K U_{\text{de}} V C_{L_\alpha}}{2W/S} \tag{55}$$

The parameter K is determined based on if the aircraft is subsonic or supersonic. For a subsonic gust,

$$K = \frac{0.88\mu}{5.3+\mu} \tag{56}$$

For a supersonic gust,

$$K = \frac{\mu^{1.03}}{6.95+\mu^{1.03}} \tag{57}$$

The mass ratio μ is defined as

$$\mu = \frac{2W/S}{\rho g \bar{c} C_{L_\alpha}} \tag{58}$$

The C_{L_α} of the aircraft was found to be 5.028 using AVL [8]. The gust vertical velocity, U_{de}, was determined using Raymer [5] and are listed in Table 23. The rough air speed V_B was determined as the intersection of the positive gust line and the maximum lift maneuver load line.

Table 23: Gust vertical velocities for different flight conditions.

	Gust velocity (ft/s)
Rough air speed (V_B)	66
Cruise speed (V_C)	50
Dive speed (V_D)	25

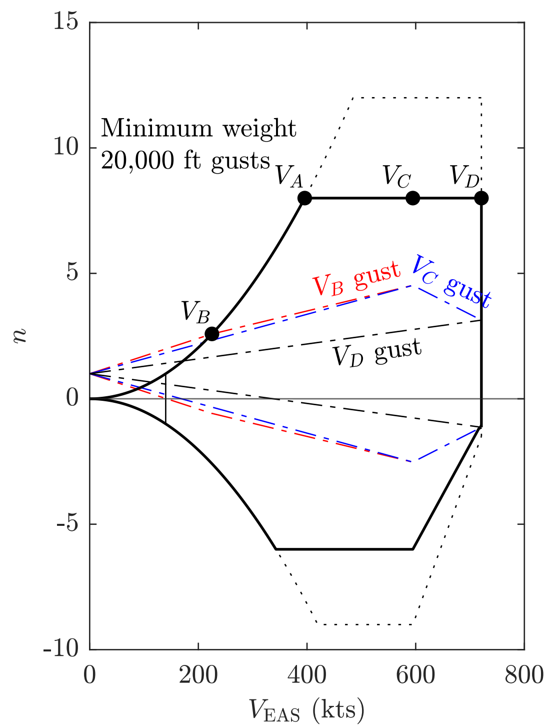

Figure 35: The V-n diagram at minimum weight. The solid black lines are the combined loads, the dotted line is the ultimate maneuver load, and the dash-dotted lines are gust loads. The 1g stall speed is denoted at the thin vertical line.

G Drag Buildup Calculations

G.1 Subsonic Parasite Drag

The component buildup method from Raymer [5] is used to estimate the subsonic parasite drag. The drag is estimated using equation 59. The components considered in the parasite drag buildup are the wing, fuselage, canopy, horizontal and vertical tails, drop tanks, and pylons.

$$(C_{D_0})_{\text{subsonic}} = \frac{\sum\left(C_{f_c}\, FF_c\, Q_c\, S_{\text{wet}_c}\right)}{S_{\text{ref}}} + C_{D_{\text{misc}}} + C_{D_{\text{L\&P}}} \tag{59}$$

The equivalent flat-plate skin friction coefficient C_f of each component is estimated using equation 60. It is assumed that the entirety of the skin has turbulent flow, as the Reynolds number is high. This assumption is also more conservative than assuming any percentage of laminar flow.

$$C_f = \frac{0.455}{log_{10}(R)^{2.58}(1 + 0.144M^2)^{0.65}} \tag{60}$$

The form factor of each of the components is estimated using equations 61, 62, and 63, which come from Raymer [5]. For the vertical tail, the FF is multiplied by 1.1 to account for the hinge line.

Wing, tail, strut, and pylon:

$$FF = \left[1 + \frac{0.6}{(x/c)_m}\left(\frac{t}{c}\right) + 100\left(\frac{t}{c}\right)^4\right]\left[1.34M^{0.18}(\cos\Lambda_m)^{0.28}\right] \tag{61}$$

Fuselage and canopy:

$$FF = \left(0.9 + \frac{5}{f^{1.5}} + \frac{f}{400}\right) \tag{62}$$

Nacelle and external store:

$$FF = 1 + (0.35/f) \tag{63}$$

The interference factor Q for each component is listed in Raymer. The interference factor is 1.05 for the tails, 1.4 for the drop tanks, and 1 otherwise.

The components included in the $C_{D,misc}$ term are the wheels, landing gear struts, and arresting hook. For the wheels and struts Raymer lists values of C_{D_π}, which is defined as the D/q divided by the frontal area.

For a combined wheel and tire, the C_{D_π} is 0.25. For a strut with Re $> 3 \times 10^5$, the C_{D_π} is 0.3. Interference is accounted for by multiplying by 1.2, and an additional 7% is added to account for landing gear doors. For a USN arresting hook, Raymer lists D/q as 0.15 ft^2. To account for leakage and protuberances such as antennas and panel edges, the $C_{D_{L\&P}}$ is approximated as 5% of the total drag, which is the upper bound of the stealth fighter estimate listed in Raymer.

G.2 Supersonic Parasite Drag

The supersonic parasite drag is estimated using equation 64, which is similar to the subsonic case discussed in §G.1.

$$(C_{D_0})_{\text{supersonic}} = \frac{\sum(C_{f_c}\, S_{\text{wet}_c})}{S_{\text{ref}}} + C_{D_{\text{misc}}} + C_{D_{\text{L\&P}}} + C_{D_{\text{wave}}} \tag{64}$$

In the skin friction calculation, the C_f is the same as in §**??**. The interference factor is not included, as it is accounted for in the $C_{D_{wave}}$ term. Thus, FF and Q are both unity. The $C_{D_{\text{misc}}}$ and $C_{D_{\text{L\&P}}}$ are calculated as in §G.1.

G.2.1 Wave Drag, $M > 1.2$

For Mach numbers above 1.2, the wave drag can be calculated using equation 65. The original value for the middle 0.2 term was 0.386, but it was changed to 0.2 based on advice from Raymer.

$$(D/q)_{\text{wave}} = E_{\text{WD}} \left[1 - 0.2\,(M - 1.2)^{0.57} \left(1 - \frac{\pi\, \Lambda_{LE\text{-deg}}^{0.77}}{100}\right)\right] (D/q)_{\text{Sears–Haack}} \tag{65}$$

The E_{WD} is the wave drag efficiency factor, and is estimated as 2.2. This is the upper bound of the historical values listed in Raymer for typical supersonic fighters.

The D/q of an equivalent Sears–Haack body, is calculated using equation 66. The A_{max} is the maximum cross-sectional area of the aircraft. If the location of maximum cross-sectional area $\ell_{A_{\text{max}}}$ is well behind the halfway point of the fuselage, the ℓ is defined as double the $\ell_{A_{\text{max}}}$.

$$(D/q)_{\text{Sears–Haack}} = \frac{9\pi}{2} \left(\frac{A_{\text{max}}}{\ell}\right)^2 \tag{66}$$

G.2.2 Wave Drag, $M_{crit} < M < 1.2$

The wave drag in the transonic regime is calculated using the graphical method described in Raymer. The M_{crit}, M_{DD}, and associated drag rise $C_{D_{\text{wave},DD}}$ are calculated using the Korn equation described in the Metabook [2]. The $C_{D_{\text{wave}}}$ at $M = 1.2$ is also found from equation 65. The $C_{D_{\text{wave}}}$ at $M = 1.05$ is assumed to be the same as at $M = 1.2$. The $C_{D_{\text{wave}}}$ at $M = 1$ is assumed to be half of that at $M = 1.05$ and $M = 1.2$, and the drag is assumed to vary linearly between $M = 1$ and $M = 1.05$. This gives sufficient fidelity to fit a spline to these points, which smoothly blends between transonic and supersonic drag.

G.3 Drag Due to Lift

The drag due to lift is calculated using historical estimations from Raymer. The subsonic value of K in the drag polar equation is calculated using equation 67, where AR is the wing aspect ratio and e is the Oswald efficiency factor. Supersonic K is calculated using equation 69. This factor accounts for both induced drag due to inviscid effects, and drag caused by flow separation due to lift. e can be estimated using equation 68 from Raymer, which is based on historical aircraft. This is preferable to using AVL [8] as this also factors in viscous drag due to lift.

$$K = \frac{1}{\pi ARe} \tag{67}$$

$$e = 4.61(1 - 0.045AR^{0.68})(\cos\Lambda_{\text{LE}})^{0.15} - 3.1 \tag{68}$$

$$K = \frac{AR(M^2 - 1)cos\Lambda_{\text{LE}}}{(4AR\sqrt{M^2 - 1}) - 2} \tag{69}$$

G.4 Flap Drag

Raymer provides two equations for calculating flap drag, one for the change in parasitic drag and one for the change in induced drag. The change in parasitic drag due to flaps is calculated using equation 70.

F_{flap} is a constant for plain flaps and slotted flaps equal to 0.0144 and 0.0074, respectively. C_f is the chord length of the flap, with C as the local chord length of the wing. S_{flapped} is the reference of the wing where the wing has a flap. The flap angle, δ_{flap}, is given in degrees.

$$\Delta C_{D_0} = F_{\text{flap}}(\frac{C_f}{C})(\frac{S_{\text{flapped}}}{S_{\text{ref}}})(\delta_{\text{flap}} - 10) \tag{70}$$

Additionally, Raymer [5] discusses a change in induced drag due to flaps. Due to difficulty in estimating the change in C_L from flap deflections, this is accounted for by estimating a Δe, a change in efficiency due to flap deflection. This is estimated using AVL to be -0.046 for full flaps, and -0.032 for half flaps. NATOPS [29] gives flap deflections for the F/A-18, which gives an estimate for the full flaps configuration.

G.5 Trim Drag

The calculation of the trim drag is difficult as the trim drag changes dramatically depending on the flight condition and the flap configuration. Therefore, the trim drag is estimated as a 5% increase in the drag due to lift. This can be found from a paper by Roskam [47], where a range of the drag due to lift increase from 0.5% to 5% is given. The upper bound of this range is selected due to the large tail area compared to the wing.

H NACA 64A006 Lift Data

Table 24: Lift Data for NACA 64A006

α	C_ℓ
-0.0459	-0.0011
0.9756	0.1142
1.9542	0.2316
2.9752	0.3556
3.9699	0.4587
4.5014	0.5296
5.0049	0.5629
5.5080	0.6052
5.9810	0.6551
6.9866	0.7533
8.0222	0.8440
9.0599	0.8865
10.0112	0.8642
11.0241	0.7937

I Additional Air Load Distributions

The lift distributions at half flaps (approach) and full flaps (takeoff) are shown in Fig 36 and Fig 37, respectively.

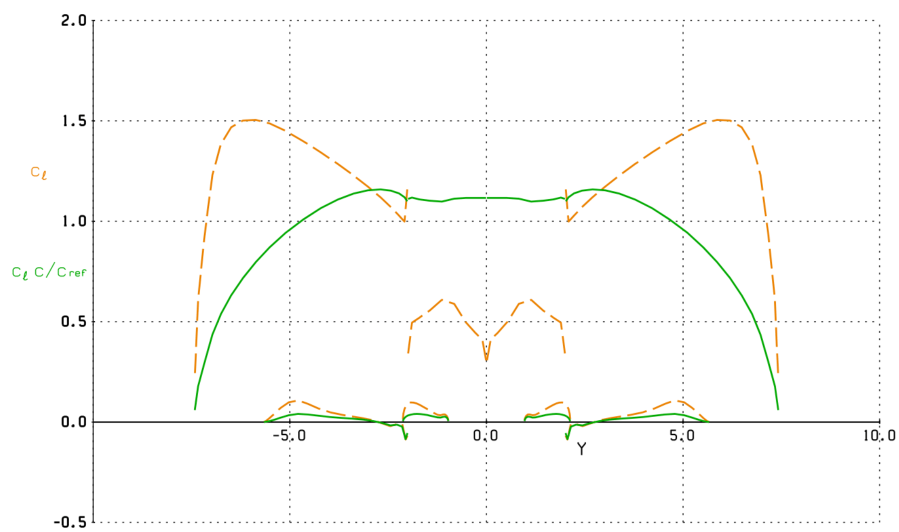

Figure 36: Half flapped spanwise lift distribution. The x-axis is wing y location in meters, and the y-axis is C_ℓ (orange) and $C_\ell \times c/c_{ref}$ (green).

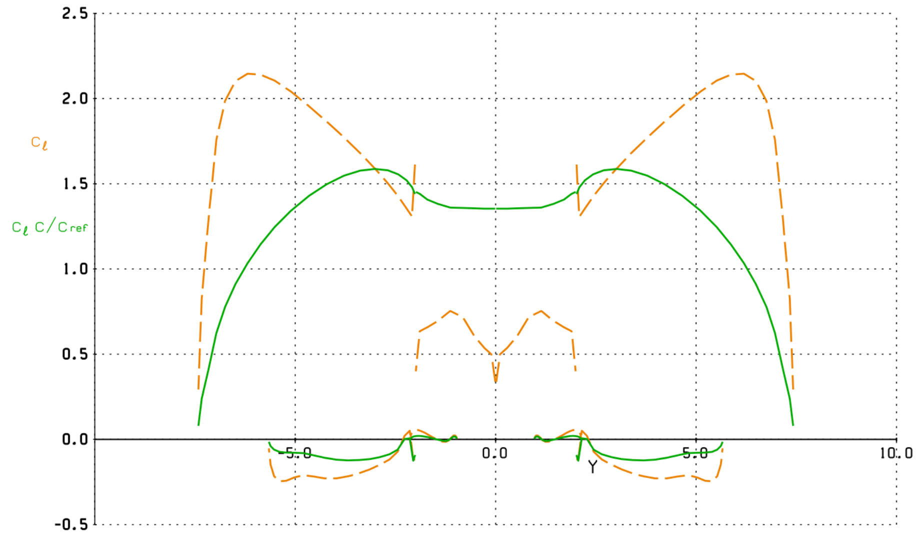

Figure 37: Full flapped spanwise lift distribution. The x-axis is wing y location in meters, and the y-axis is C_ℓ (orange) and $C_\ell \times c/c_{ref}$ (green).

J Mission Fuel Refinement

The weight fractions for cruise and climb mission segments were refined using a multi-segment approach.

J.1 Cruise Weight Fraction

Cruise mission segments are modeled as a cruise climb over a fixed distance. The distance traveled is subdivided into n sub-segments, with each assume to be flown at a constant altitude. The entire cruise is flown at a fixed Mach number of 0.84. The range traveled during a cruise subsegment is derived as follows.

$$R = \int_{W_{i+1}}^{W_i} \frac{V}{c} \frac{L}{D} \frac{dW}{W}$$

Velocity is substituted and L/D is replaced by their respective coefficients.

$$R = \int_{W_{i+1}}^{W_i} \frac{1}{c} \sqrt{\frac{2W}{\rho S C_L}} \frac{C_L}{C_d} \frac{dW}{W}$$

Integrating yields equation (71). The equation can be rearranged to find the subsegment weight fraction W_{i+1}/W_i.

$$R = \frac{2}{c} \sqrt{\frac{2}{\rho S}} \frac{C_L^{1/2}}{C_D} \left(\sqrt{W_i} - \sqrt{W_{i+1}} \right) \tag{71}$$

Since the cruise segment is divided into sub-segments, the range R, thrust-specific fuel consumption c, and wing reference area S, are known. The weight at the start of the cruise segment W_i, or sub-segment is also known. To reduce fuel burn during a sub-segment, $\left(\sqrt{W_i} - \sqrt{W_{i+1}}\right)$ is minimized.

This can be done by maximizing all other unknowns on the right hand side of equation (71). For a given W_i, the altitude that maximizes the expression in equation (72) is found using MATLAB's `fmincon`. This is determined at a fixed cruise Mach number of 0.84, and for a given aircraft (i.e., for some S, C_{D0}, etc.).

$$\frac{1}{\rho S}\frac{C_L^{1/2}}{C_D} \tag{72}$$

The assumption is made that the aircraft's velocity does not change significantly during the cruise sub-segment, and that the change in altitude during the sub-segment is small. This holds as long as the change in weight is small. The weight fraction for the entire cruise segment is the product of the weight fractions for all sub-segments as shown in equation (73)

$$\text{Weight Fraction} = \prod_{i=1}^{n} \frac{W_{i+1}}{W_i} \tag{73}$$

J.2 Climb Weight Fraction

Climb mission segments follow a similar sub-segment procedure as the cruise mission segment. Instead of subdividing a given range, altitude is subdivided into n sub-segments. Sub-segment weight fraction is determine using the energy method.

$$\frac{W_{i+1}}{W_i} = \exp \frac{-c\Delta h_e}{V(1-D/T)} \tag{74}$$

h_e in equation (74) is the energy height which is defined as

$$h_e = h + \frac{V^2}{2g} \tag{75}$$

The velocity flown the climb segment is the velocity for the best rate-of-climb which is found using equation (76) from Raymer [5]. Lift is assumed to be equal to weight for small climb angles, which can be used to calculate drag. Wing loading is calculated for the beginning of a sub-segment. The thrust-to-weight ratio and the thrust must be determined numerically since the aircraft's velocity influences thrust lapse, which changes thrust-to-weight, which changes the velocity for best rate-of-climb. The change in velocity as the altitude changes is assumed to be small.

$$V = \sqrt{\frac{W/S}{3\rho C_{D0}}\left(\frac{T}{W} + \sqrt{\left(\frac{T}{W}\right)^2 + 12C_{D0}K}\right)} \tag{76}$$

The weight fraction for the entire climb segment is determined using equation (73). Credit is also taken for the distance traveled during the climb. This is done by calculating the specific excess power using equation (77).

$$P_s = \frac{TV - DV}{W_i} \tag{77}$$

Dividing the change in energy height by the specific excess power yields the time taken to fly a sub-segment. Assuming a small flight-path angle, multiplying by V yields the distance traveled for a sub-segment as shown in equation (78).

$$\Delta x = \frac{\Delta h_e}{P_s} V \tag{78}$$

An additional consideration is the determination of the climb height. The climb should transition to a cruise segment at the altitude where the aircraft weight at the end of the climb yields an identical altitude to begin an optimal cruise.